

Remote Bushfire Information E-Display

ELEC3117 Final Report 26T2



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1. Executive Summary

The Remote Information E-Display is an autonomous information system designed to improve the availability and accessibility of safety information in remote national park trails and other areas where conventional signage is difficult to update or maintain. The system addresses limitations associated with static signs by providing remotely updateable information without requiring grid power or internet connectivity at the display location.

The system was designed around several key requirements. This includes autonomous solar charging, radio communication, low power operation of components, automated information updates, outdoor protection and infrequent maintenance. The display is intended to communicate more than simply a fire danger rating, with information including trail status, fire bans and distance and location of emergency assembly points. A simplistic and inexpensive design is used to allow for nationwide deployment including remote areas and areas of different wealth.

Existing autonomous solar-powered fire danger signs demonstrate safety information in remote locations. However, the Remote Information E-Display extends this concept by providing multiple categories of safety information through a single remotely updateable display. Its strategic advantages include the ability to remotely update safety information, operate independently of internet and power infrastructure, and potentially offer a lower-cost solution for deployment across multiple remote locations.

The current prototype demonstrates the core system required to achieve these objectives, including solar-powered operation, wireless data transmission, remote information processing and digital display. Further development is required to move the prototype into a market ready product. including long-term environmental protection, power optimisation, manufacturing, reliability and cost reduction. Future development will therefore focus on improving the manufacturability of the system while maintaining its operation.

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2. Project Description

2.1 Problem Statement

2.1.1 Users

Our product is designed for hikers and other outdoor recreationalists (campers, trail runners, etc.), supporting fire service authorities such as the RFS, and park rangers in remote and/or bushfire-prone areas.

Hikers and outdoor recreationalists are considered as ‘users’, who will view the product. Park rangers and fire service authorities are considered as ‘operators’, who will purchase and operate (maintain) the product. Both groups are to benefit from this product.

2.1.2 Necessity

Australia notably has some of the most widespread bushfires globally, each wildfire burning 909 hectares on average in 2025 (Global Wildfire Information System (2026) – with minor processing by Our World in Data). This has led to hundreds of deaths across the country, causing 33 direct and 417 indirect deaths purely in the Black Summer 2020-2021 bushfires (“Black Summer,” 2023; Borchers Arriagada et al., 2020).

Specifically, Australia’s remote, bushfire-prone areas pose an even larger threat to life, as the 90th percentile of emergency response times take up to 36 minutes compared to the 13.2 minutes residentially (“D Emergency Management - Report on Government Services 2026,” 2026).

This is compounded with the low bushfire awareness and safety precautions taken when hiking in Australia’s bushfire-prone areas, with 40% of hikers not checking the local fire danger rating before entering a trail (Kuru Footwear, 2025). Furthermore, over 50% of people residentially don’t know their nearest fire assembly point, with higher percentages projected for areas that people are not familiar with (McLeod, 2025). Notably, 69% of Australia’s bushfires between 2019-2020 are considered to be man-made, proving a need for total fire bans to be known (“Fire 2021 | NSW State of the Environment,” n.d.).

An extra vulnerability within Australia’s remote areas is connectivity and access to emergency information and instructions, with 1927 facility outages purely within the 2019-2020 Sydney storms (“Impacts of the 2019-20 bushfires on the telecommunications network”, 2019). These remote areas significantly lack public infrastructure and are often off-grid.

2.1.3 Product Functionality

The proposed solution is an autonomously updated, solar-powered off-grid bushfire information E-display. It consists of a long range (LoRa) wireless receiver to update safety information on a display positioned at the entrance to any hiking trail. It is completely self-sustaining, as it can receive updates at any time and self-charge via solar power, thus having no reliance on the energy grid or any telecommunications networks. The safety information involves:

- Fire danger rating
- Distance and direction to the nearest fire assembly point
- Total fire bans if any are active
- Trail closed signage if necessary

A main server will be used to receive information and update receivers at any time.

2.1.4 Comparisons With Existing Products

The two existing products considered are:

- Static signs
- HVIS LED Fire Danger Rating signs

The features examined are:

- Live updates: Information must be up-to-date to be useful to outdoor recreationalists
- Deployment cost: Must be low-cost in order to be deployed nation-wide
- Maintenance cost: Same as above
- Grid independence: Necessary for remote deployment and usage during outages
- Network independence: Same as above
- Visibility at night: Information must be clear at all times for safety

TABLE I

COMPARISON WITH EXISTING PRODUCTS

Features	Static Signs	HVIS LED Signs	Remote Bushfire Information E-Display
Live updates	None	Only if connected to network	As long as main server is functional
Deployment cost	Small (< \$100)	Moderate (~\$500-\$1000)	Small (< \$100)
Maintenance cost	None	None if updated via network. Moderate if updated manually.	None
Grid independence	Yes	Yes	Yes
Network independence	Yes	No, unless updated manually.	WiFi, satellite and cellular independence. Uses radio (LoRa).
Visibility at night	No	Yes	Yes

From Table I, the proposed solution succeeds in matching all the features listed, whilst the two existing solutions each have their own upsides and downsides.

Notably, static signs are completely unable to update with live information unless a worker manually updates it. This would considerably increase maintenance costs, especially considering travel to remote areas, and still result in significant delays if not updated multiple times a day. Furthermore, static signs lack visibility required for low-vis environments. Some solutions may consider an external light, however in off-grid areas this would incur a significant cost purely for a solar-powered light.

HVIS LED signs are high quality, however are significantly more expensive due to their material and size, and have a dependence on WiFi or cellular connectivity (unless updated manually, increasing maintenance costs). Satellite may be used, however this also incurs additional costs for usage. Due to

their cost, they cannot be conceivably deployed nation-wide, especially in poorer remote areas. They also still lack the necessary information that the proposed solution provides, since they only provide a fire danger rating.

The proposed solution succeeds in every category, only having a dependence on a main server that is relied upon to upload the information. Since it operates using LoRa/radio, it is completely independent of any established networks. It is self-powered via solar, similarly to the HVIS LED signs, however remains cheap for nation-wide deployment.

2.1.5 Key Challenges

2.1.5.1 Long-Range Wireless Communication

One of the most important decisions for this product was the wireless communication it uses. WiFi and possibly bluetooth connectivity is considered as the simplest method to transmit information. Cellular is also a highly viable option. All of these are simple to implement and offer high bandwidth, and can be done on any wireless-supporting microcontroller. However, they have significantly limited range. Satellite is highly difficult to implement, and would incur a constant cost to maintain. LoRa is also complex and incurs additional cost to support.

This challenge leads to the decision to use LoRa, which requires extra complexity with an external module and sending device, but boasts a much larger range that is more suitable for remote deployment.

2.1.5.2 Efficiency & Self-Powering

In order to be completely independent from the grid, the device needs to be self-powering at a level where charging can both outpace discharge, as well as last long periods without charge (e.g. overcast for multiple days). This requires the use of more efficient power stages, devices, and features in order to conserve as much power as possible. This also strikes a balance between high-capacity, expensive devices such as large solar panels and battery packs, and lower-capacity but cheaper devices.

This leads to the decision of using a small solar panel and 2S Li-Ion configuration, along with a motion-detecting waking system.

2.1.5.3 Automatic Waking System

In order to not drain power continuously when the display is not required to stay on, a system is required to check when users (outdoor recreationalists) pass by, and activate the display accordingly. The passive power draw, even when the display is off, must also be optimised via MCU sleep features.

2.2 Concepts

2.2.1 Product Concept/User Needs

In order to improve safety for Australian hikers in remote areas, the product must aim to:

- Be suitable for deployment at entrances to remote hiking trails and other public outdoor areas in Australia
 - This requires a cheap product in order to be deployed in poorer areas, specifically < \$100.

- The receiver should maintain a normal operation for at least 7 days without solar charging under the worst case operating conditions. The wireless communication system should also transmit information over a minimum distance of 2km.
- Display fire danger ratings, nearest fire assembly points, total fire bans and trail status.
 - Some sort of display mechanism must be used to communicate this information in an easily understandable format. The display should use an appropriate visual ranking and colour scheme to distinguish different levels of fire danger. This should be consistent with relevant and pre-existing fire danger rating systems.
- Be visible at all times.
 - The display should be easily readable at all times, in all weather conditions including daytime, nighttime and rain.
- Use live information.
 - The product must always have up-to-date information to prevent giving outdated information indicating trails are safe to enter when they currently are not. Also allows authorities to close trails or enact fire bans at any given moment. Information must update on the display in less than 10 seconds from transmission.
 - Information must be transmittable in a simple manner. A trained operator should be able to transmit an information update to the display within 60 seconds.
- Require no interactivity.
 - The product should give the safety information indiscriminately, without any required inputs from outdoor recreationalists (e.g. no buttons, no touchscreen, no required apps or subscriptions).
 - Park rangers and emergency service workers should not need to travel to the receiver in order to operate it. All information must be able to be transmitted wirelessly. Any maintenance or servicing/cleaning should be at most every year (e.g. solar panel cleaning, battery replacement, etc.).
- Continue operation without constant updates.
 - The product must continue to display the last update transmitted to it until it receives another update. The product must not request information and crash or stop displaying if none is received. This allows continued operation even if a transmission is not sent for a significant amount of time.
- Operate in remote conditions.
 - The enclosure should provide a minimum IP43 level of protection against solid objects greater than 1mm and water spray up to 60 degrees from vertical.

2.2.2 Design Concept

Fig. 1 demonstrates the product's high level diagram, which feasibly implements a solution to all the given user needs. It has an abstract level of each component, which provides the minimal specification required to meet these needs. Specifically, it has no inputs for the outdoor recreationalists to interact with, and uses a backlit display to remain visible at all times.

A motion sensor connected to the microcontroller prompts it to turn on the display only when necessary (a passerby trips the sensor), conserving power by having the display off whenever possible, yet still being visible for any users. The long-range receiver receives radio transmissions and provides the transmission information to the microcontroller, prompting it to update the display with the new details. Notably, only the receiver can initiate updates, so the display and microcontroller will continue operating without needing constant updates. All bushfire safety information (fire danger

rating, nearest fire assembly point, total fire bans, trail status) can be uploaded to the wireless transmitter, ensuring that live information can be displayed.

The solar panel charges the battery via a BMS whenever possible, ensuring the product requires no maintenance in the form of charging. The batteries are then dropped to 3.3V in order to supply the correct voltage for each component.

With a reasonably sized display and solar panel, the product can be kept well under \$100 in order to be deployed nation-wide at every hiking trail. Lastly, by placing the product in a 3D printed enclosure, the product can be reasonably waterproof and endure wet weather conditions for long periods of time.

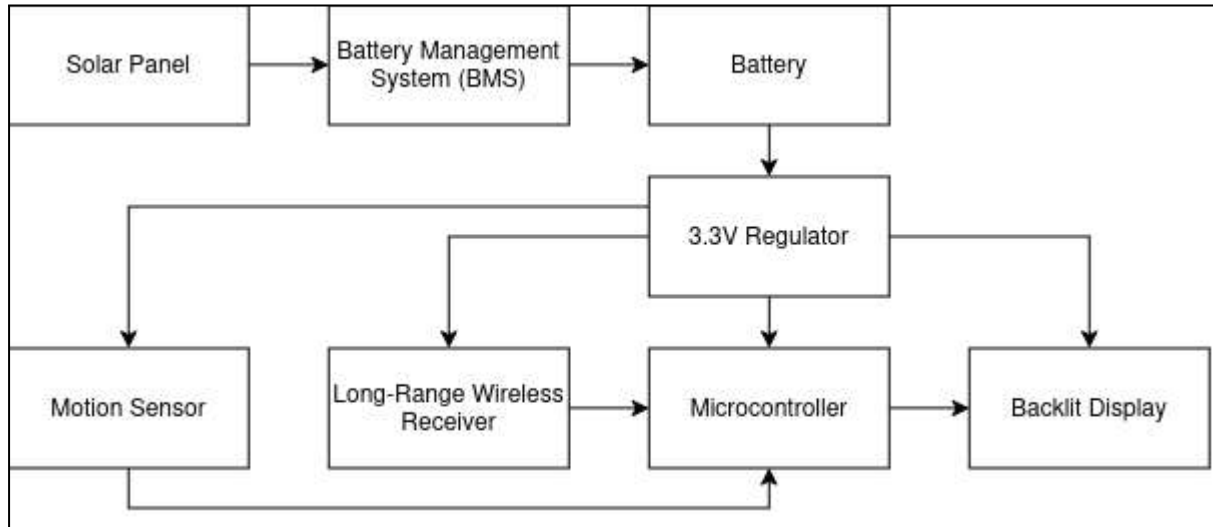


Fig. 1. High Level Diagram.

2.3 Unique Selling Points

Note that this section considers the final design instead of just the high-level design in Fig. 1.

2.3.1 Long Range Communication

Utilises a long-range (LoRa) wireless receiver to allow for transmissions over 2km in range via radio, which also penetrates through forested areas. This outperforms all existing solutions since they use WiFi, cellular or satellite communications. This allows for operators to send updates to remote areas without the cost of traveling to the deployed area. It also allows information to be displayed to outdoor recreationalists in areas without any network connection.

2.3.2 Motion Activated Power Management

Utilises a motion sensor that ensures the display only turns on when a user is nearby. This conserves power when nobody is nearby by turning off the display and putting the MCU to sleep (with an interrupt connected to the motion sensor pin). This allows the product to stay long-lasting without operators having to pay the cost for large solar panels or batteries. It also allows for users to still view bushfire safety information even after long periods of overcast or rain.

2.3.3 Efficient Power Architecture

Through the use of a buck converter *and* an LDO, the product benefits from a more efficient power stage compared to a single LDO. This increases the battery lifetime for the product, and similarly to

2.3.2, ensures efficient power use so that operators don't need to pay for larger panels or batteries, and so users can view bushfire information after long overcasts without solar charging.

2.3.4 Localised Information

Due to the marginal cost of the product compared to alternate solutions such as the HVIS LED fire danger rating signs, this product can be deployed nationwide across all hiking trails. This allows for each deployed product to have more localised information, including more accurate fire danger ratings for the given location and the ability to notify of total fire bans in a more accurate region. This gives users more accurate information pertaining to the trail they are entering, and allows them to make more educated decisions. It also improves the safety of users, as more accurate information prevents generalisations from being made about safety across a region.

2.3.5 Improved Safety Information

Other solutions only display small amounts of information, most notably the fire danger rating. The proposed product benefits the users' safety by also incorporating information on the nearest fire assembly point, total fire bans, and trail status. This also benefits operators through the ability to share information on total fire bans, lowering the amount of man-made bushfires.

2.3.6 Lightweight Installation

In an effort to make the product accessible nationwide, it also boasts a very simple deployment which can be done by a single operator. HVIS LED fire danger ratings signs are significantly heavier, are required to be placed on high altitudes for visibility, and thus require machinery to be installed. The proposed product can simply be placed on a wooden pole, and can easily ($< 10\text{kg}$) be carried by a single operator.

2.3.7 Low Thermal Stress

Compared to HVIS LED fire danger rating signs, the proposed solution utilises a motion sensor working system and a low-power display to significantly reduce operating heat. This is especially useful in Australia's hot summers, which can reach up to 40°C and lead to lower battery lifetimes. This benefits operators by reducing the amount of maintenance from battery replacements, reducing the cost of implementation.

2.4 Product Requirements

2.4.1 Battery Longevity

The product must theoretically power a display with backlight ON for ≥ 10 hours. That is, the current draw measured from the battery with backlight ON (in mA) must satisfy $i \leq \frac{\text{Capacity}}{10}$, where capacity is in mAh.

2.4.2 Wireless Transmission Range

The device must be able to receive information 2km away from a LoRa transmitter, and correctly update the display in $< 10\text{s}$. The transmission must correctly carry all relevant data (fire rating, distance + direction to fire assembly point, trail status, fire ban status).

2.4.3 Motion Sensor Wakeup

The device must awake from sleep and correctly turn on the display $< 500\text{ms}$ after a HIGH output from the motion sensor.

2.4.4 Motion Sensor Activation

The motion sensor must output HIGH when a person passes by $< 3m$ away from the sensor.

2.4.5 Display Details

The display must provide fire danger rating following AFDRS (None, Moderate, High, Extreme, Catastrophic) in text, distance to the nearest fire assembly point in metres, direction to the nearest fire assembly point in the secondary intercardinal direction in text and with an arrow, total fire ban (if true) in text, and trail closed status (if true) in text.

2.4.6 Operating Voltages

The 7.4V-5V buck converter must be able to take a range of 6V-8.4V (discharged to full charged) and output $5V \pm 0.2V$. The LDO must output $3.3V \pm 0.1V$ at all times.

2.4.7 Sleep Mode

The device must go to sleep when the motion sensor has not outputted HIGH for $5 - 15s$. During sleep, the display backlight must be OFF and the MCU must enter sleep mode, which is considered successful if the device draws $< 100\mu A$.

2.4.8 IP43 Operation

The device must operate as IP43, meaning it cannot have solids $< 1mm$ damage it, and water at angles $< 60^\circ$ damage it. Damage involves any dysfunction or missed requirements that would otherwise not occur.

2.4.9 Unit Cost

The total cost per receiver unit must be $< \$150$ excluding shipping. This can consider bulk prices up to 1000 units.

2.4.10 Display Attention

The display must use colour or movement when activated, or any equivalent to get user attention as they pass by. This includes use of flashing, visual rankings (e.g. red for danger) or any other animation.

2.4.11 Display Visibility

The display must be visible in all conditions, including fog, rain, nighttime or daytime, to users passing by within 3m of the device.

2.4.12 Ease of Use

Operators must be able to transmit bushfire safety data from their location within 60 seconds.

3. Product Overview

3.1 System Overview

The Remote Information E-Display is a wireless information display designed to provide up-to-date safety information in remote or difficult-to-access locations for national parks. The system allows information such as bushfire risk, fire-ban status, trail closure status, direction to a relevant hazard and distance information to be transmitted wirelessly to a remote display. This removes the need for information to be manually updated at the display location.

The final prototype consists of three major subsystems: the Power Module, Sender Module and the Receiver Module. A high-level representation of these subsystems and their interfaces is shown in Fig. 2.

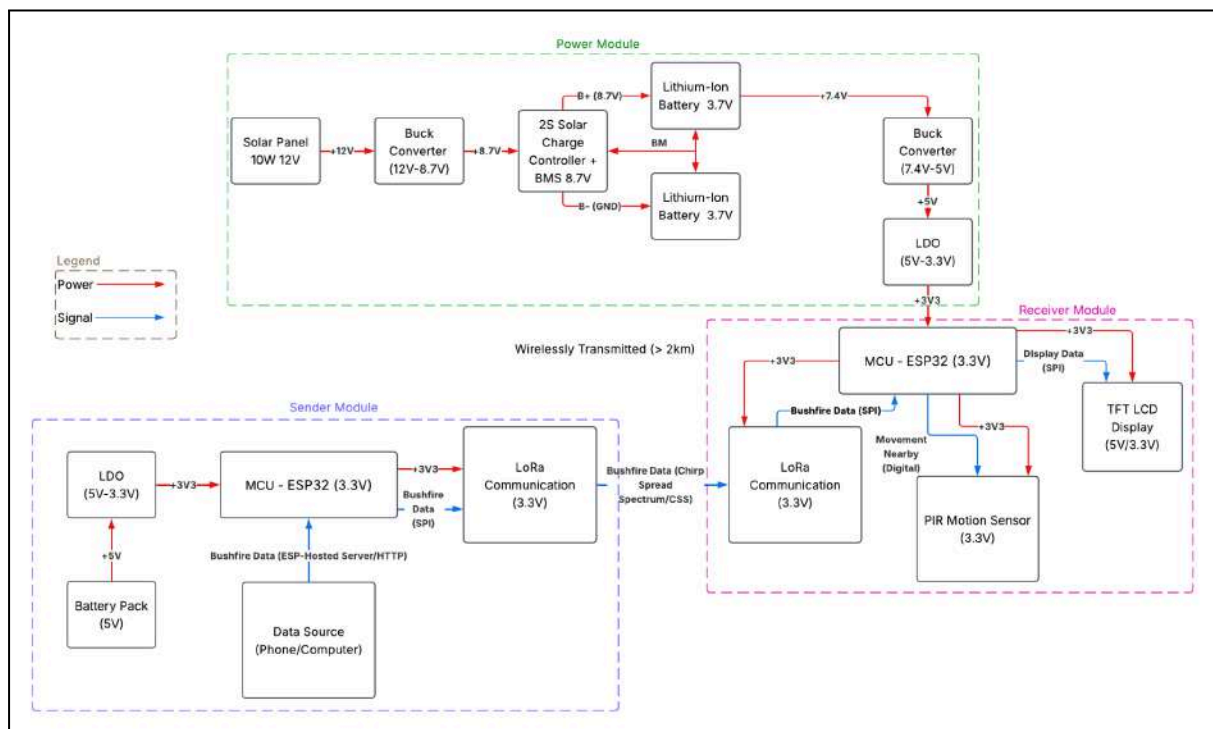


Fig. 2. Subsystem Diagram.

The Power Module supplies power to the system. A 10W, 12V solar panel provides the primary source of energy. This is regulated to a two cell lithium-ion battery charging and battery-management system. The 2S battery configuration provides a nominal 7.4 V supply to the power converter. This voltage is converted to 5 V and then 3.3 V for the receiver electronics.

The Sender Module is responsible for receiving and transmitting the information that is displayed to the user. A data source, such as a mobile phone or computer, communicates with the ESP32 through an ESP-hosted HTTP server. The ESP32 processes this information and communicates with the LoRa transceiver using SPI. The LoRa module then wirelessly transmits the information to the remote receiver.

The Receiver Module contains a second ESP32 and LoRa transceiver. The LoRa module receives the transmitted information and communicates it to the ESP32 using SPI. The ESP32 then processes the received information and sends the appropriate display data to the TFT LCD through another SPI interface. A PIR motion sensor is also connected to the receiver ESP32 to detect nearby movement and control the display based on user presence.

3.2 Major System Interfaces

The major interfaces between the system subsystems are shown in Table II. These can be divided into power, communication, and sensing interfaces.

TABLE II

SYSTEM INTERFACES

Interface	Type	Purpose
12V Solar Panel -> Buck Converter (12V - 8.7V)	Power	Reduces the solar panel voltage to a suitable charging voltage
Buch Converter -> 2S Charge Controller / BMS (8.7V)	Power	Supplies power for battery charging
2S Battery -> Buck Converter (7.4V - 5V)	Power	Provides stored energy to the receiver system
Buck Converter (5V)r -> LDO (5V - 3.3V)	Power	Produces a 5V supply
LDO -> ESP32/LoRa/PIR	Power	Provides a 3.3V supply
Data Source -> Sender ESP32	Communication	Transfers bushfire and safety information to the sender
Sender ESP32 -> LoRa	SPI	Transfers bushfire and safety information to the sender
Sender LoRa -> Receiver LoRa	Wireless	Transmits information over the long range wireless link
Receiver LoRa -> Receiver ESP32	SPI	Transfers received bushfire data to the receiver
Receiver ESP32 -> TFT LCD	SPI	Sends display information to the LCD
PIR -> Receiver ESP32	Digital control / sensing	Detects nearby movement

The system uses SPI for local peripheral communication, while LoRa provides the long-range wireless connection between the sender and receiver. The diagram identifies the intended wireless communication distance as greater than 2 km.

3.3 Principle Of Operation

The principle of operation can be divided into five stages: information generation, data transmission, wireless connectivity, data processing and information display.

3.3.1 Information Generation

The system begins with a data source. This is represented in the prototype by a phone or computer. The data source provides the information that is intended to be displayed remotely. This includes the relevant bushfire and safety information.

The sender ESP32 hosts a local HTTP server and receives the information from the data source. This would allow the information to be updated without in person access to the remote display.

3.3.2 Data Transmission

Once received by the sender ESP32, the data is put into the system. The ESP32 communicates this information to the sender LoRa module through an SPI interface. The LoRa module then converts the information into a wireless transmission and sends it over the long range communication to the remote receiver.

3.3.3 Wireless Connectivity

The LoRa transceiver in the Receiver Module continuously monitors for incoming transmissions. When a valid transmission is received, the receiver LoRa module transfers the received information to the receiver ESP32 using SPI

3.3.4 Data Processing

The receiver ESP32 processes the information received from the LoRa module and determines the information that needs to be presented to the user. The ESP32

3.3.5 Information Display

The processed information is transmitted from the receiver ESP32 to the TFT LCD through an SPI interface. The TFT LCD then presents the received safety information visually to users at the remote location.

The resulting data flow through the system can therefore be summarised as:

Data Source -> Sender ESP32 -> SPI -> Sender LoRa -> Wireless LoRa Link -> Receiver LoRa -> SPI -> Receiver ESP32 -> SPI -> TFT LCD

This architecture allows information to be generated and updated remotely before being transmitted to and displayed at a physically separated location.

3.4 System Inputs & Outputs

The main inputs and outputs of the product are summarised in Table III:

TABLE III

INPUTS & OUTPUTS

System	Inputs	Outputs
Sender Module	Bushfire/safety information from phone or computer	LoRa wireless data
LoRa communication link	Wireless data from sender	Wireless data to receiver
Receiver Module	LoRa data, PIR movement	Display data

	signal	
TFT Display	SPI display data	Visual safety information to users
Power Module	Solar energy	Electrical Power
PIR Sensor	Nearby movement	Digital movement signal

The primary system input is the safety information supplied by the data source, while the primary system output is the information presented visually to the user on the TFT display. Solar energy provides the primary energy input for the remote receiver system.

3.5 Relationship To The Design Concept

The final Remote Information E-Display was designed to address the requirements and constraints established during the initial design process. The primary design objectives were to provide an updateable information display that could operate independently of grid power and internet connectivity, require infrequent maintenance and remain suitable for deployment in outdoor environments.

3.5.1 Power & Communication

The remote system is powered by a solar panel and lithium ion battery storage. The solar charging system provides a mean of replenishing the battery during operation hence reducing the need for manual intervention

3.5.2 Low Power Operation

Low power consumption was an important design consideration because the remote display is powered by a solar energy supply. The design therefore incorporates low power components and a PIR motion sensor to detect when a user is near the display.

3.5.3 Automated Information Updates

The product was designed to eliminate the need for rangers to manually update information at the display. The sender module receives information from the data source and transmits it to the remote receiver using the LoRa communication link.

This allows information presented on the display to be changed without physically accessing the sign.

3.5.4 Outdoor Deployment

The product is intended for deployment in outdoor environments such as national parks, hiking trails and other remote public areas. A waterproof enclosure is therefore added into the design to protect the electronic components from environmental conditions.

Weatherproofing the box was a key system constraint because the product is intended to remain outdoors for extended periods while requiring maintenance no more frequently than a couple of years.

3.5.5 User Information & Symbolology

Another design objective was to provide information that can be understood by a broad range of users, including people who may have limited English proficiency. The use of symbols provides a

language-independent method of communicating important information and is particularly relevant for culturally and linguistically diverse (CALD) users. In addition to bushfire information, the proposed information set includes trail status, fire-ban information and assembly-point information.

3.5.6 Comparison With Existing Solutions

Existing products, such as autonomous solar-powered LED fire danger rating signs, already demonstrate that solar charging and autonomous operation can be applied to outdoor safety signage. However, these solutions primarily focus on communicating fire danger ratings.

The remote Information E-Display expands upon this concept by combining autonomous power and wireless communication. Its key differentiating features are:

Symbology: To improve accessibility for no English speaking and CALD users

Trail Status: To inform hikers whether a trail is open or closed

Fire Ban Information: To communicate bushfire safety information

Assembly Point Information: To instruct users with relevant emergency information

Lower Cost Design: Providing a more affordable alternative for deployment across remote locations.

These features would shift the product from a fire danger sign towards a multi purpose remote safety information display.

4. Detailed Design

4.1 Subsystems

4.1.1 Receiver Power

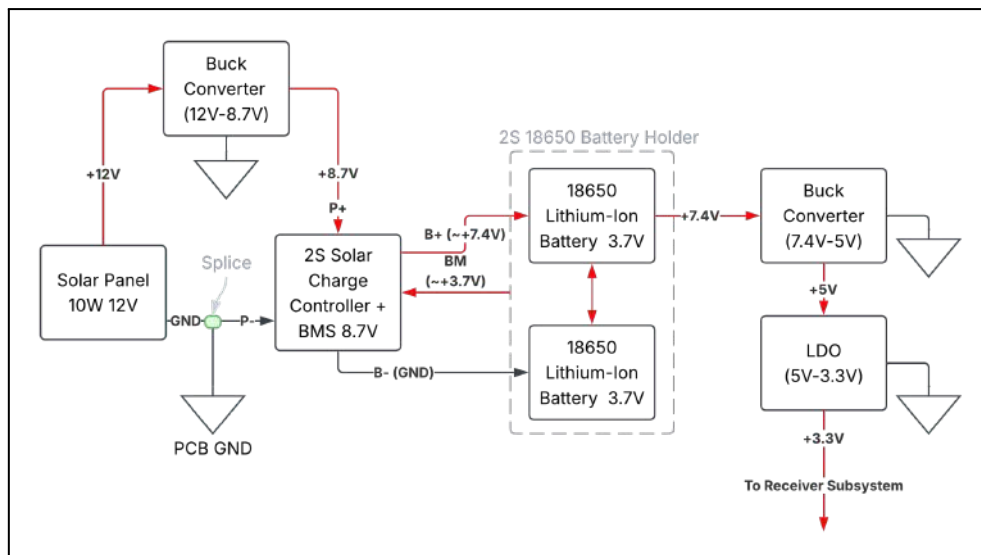


Fig. 3. Receiver Power Subsystem

The receiver power subsystem (Fig. 3) generates, stores and regulates the energy to be delivered to the receiver's core module (MCU, display, LoRa, motion sensor). It consists of:

- 10W 12V Polycrystalline Silicon Solar Panel (110mm x 136mm)
- 2x LMR51610XFQDBVRQ1 Texas Instruments Buck Converter (12V to 8.7V, 7.4V to 5V)
- 2S 10A Battery Protection Board (BMS) (47mm x 23mm)
- 2S 18650 Battery Holder
- 2x 18650 3.7V Li-Ion Batteries
- TLV73333PQDBVRQ1 Texas Instruments LDO (5V to 3.3V)

Operation

The system generates energy via a small solar panel mounted to the top of the enclosure, which produces a voltage up to 12V in full daylight and can support up to 10W.

Notably, the chosen BMS does *not* take 12V as an input, and instead requires 8.4V-9V. This requires the use of a buck converter to drop down to ~8.7V whilst maintaining maximal efficiency. Thus, the solar panel's GND is soldered onto BMS GND (P-) (and also spliced into the PCB GND) and its 12V power connects to a jumper pin on the PCB (via jumper cable) to drop down to 8.7V. This 8.7V output connects to another jumper pin that connects to the BMS PWR (P+) via a jumper cable soldered onto the BMS.

The BMS is vital to the subsystem to ensure the energy given to the 2S Li-Ion battery configuration is regulated, preventing overcharge, overdischarge and overheating. It maintains the charge of the 2S Li-Ion battery configuration, using an extra BM (Battery Middle) input to make sure both batteries are of similar voltage.

The two batteries connect back to the PCB using a jumper cable, supplying a nominal 7.4V to the first buck stage, dropping efficiently to 5V before a 3.3V LDO for clean power to the receiver subsystem.

The buck is chosen as the first stage to conserve power due to its high efficiency, and the LDO is required for the sensitive MCU and LoRa module, giving a cleaner 3.3V than any buck would provide.

4.1.2 Receiver Module

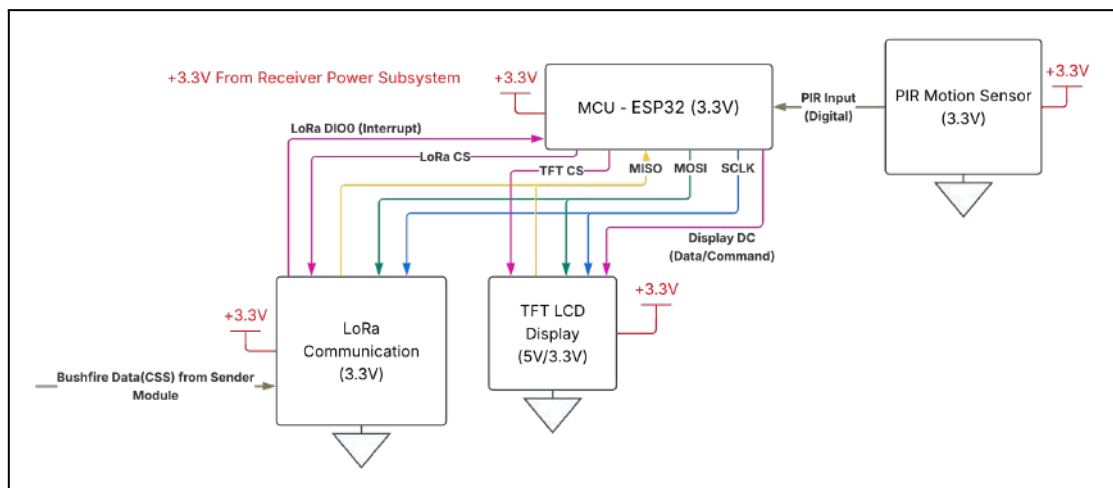


Fig. 4. Receiver Module Subsystem.

The receiver module subsystem (Fig. 4) wirelessly receives and displays bushfire data with a motion-activated display. It consists of:

- ESP32-WROOM-32E-N8 (MCU)
- Ra-02 SX1278 (LoRa Module)
- Marhynchus ILI9488 3.5" LCD Display (480 x 320)
- AM312 PIR Motion Sensor

Operation

The system is operated by the ESP32 chip, which acts as a master to the SPI communication bus with the TFT LCD display and LoRa module. It also receives the motion sensor input via a digital pin, pulled HIGH when motion is detected and LOW otherwise.

The LoRa module receives communications via radio using the CSS (Chirp Spread Spectrum) encoding, where increases in frequency and decreases in frequency indicate 1s and 0s respectively. These communication signals come from the sender's LoRa module in order to share new bushfire safety data.

When a transmission is received, the receiver's LoRa module uses DIO0 to create an interrupt, which the ESP receives and promptly retrieves the data via SPI, pulling the LoRa CS LOW..

Using the same communication bus, the ESP pulls the LoRa CS back to HIGH and pulls the display CS HIGH. It then resets the screen and prompts the display to draw with all the new data. The display receives this information with the display driver, an ILI9488 chip that executes all prompted actions. This refreshes the screen with all of the new bushfire safety data.

Lastly, the motion sensor is read as an input to the ESP, and is used to decide when to turn the display on and off. The ESP's software dictates it to turn off the display's backlight via SPI when the motion sensor has not pulled HIGH for 10 seconds. This is called the 'sleep timer', and is configurable in code.

4.1.3 Sender Module

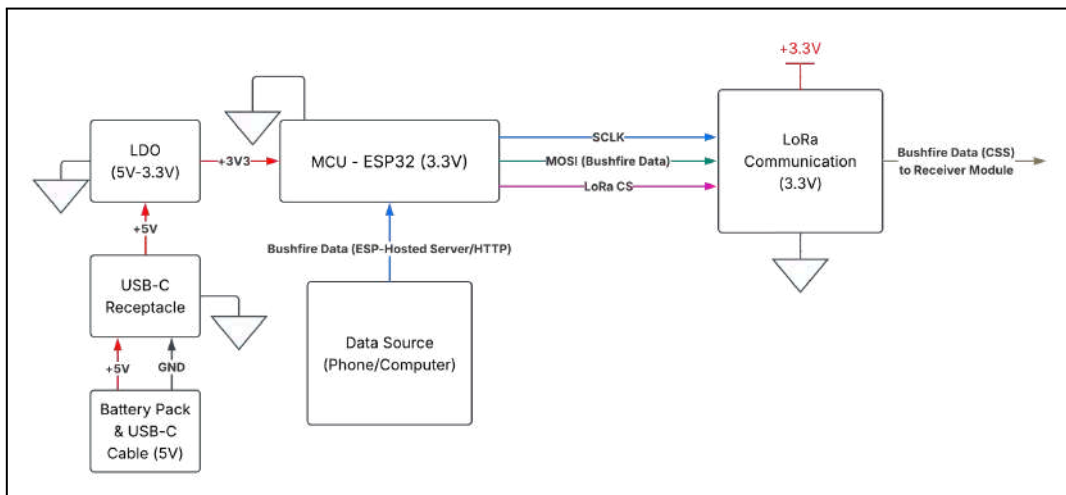


Fig. 5. Sender Module Subsystem.

The sender module subsystem (Fig. 5) receives bushfire safety data wirelessly over WiFi from a personal device, and communicates the data via SPI to a LoRa module to broadcast. It is powered by any 5V power source, e.g. a battery pack. It consists of:

- ESP32-WROOM-32E-N8 (MCU)
- USB-C Receptacle
- TLV73333PQDBVRQ1 Texas Instruments LDO (5V to 3.3V)
- Ra-02 SX1278 (LoRa Module)
- Any 5V+ USB-C power source and any device with WiFi and browser.

Operation

The system receives power by any 5V+ power source that can transmit power over USB-C, such as a battery pack or laptop. This supplies 5V power and ground to the PCB which is received with a USB-C receptacle, and is dropped via an LDO to a clean 3.3V for the sensitive MCU and LoRa module.

Bushfire Warning Control

Fire Rating

Catastrophic ▾

☒ Trail Closed

☒ Total Fire Ban

Fire Direction

South-East ▾

Distance To Assembly Area

150 metres

Broadcast Sign

Fig. 6. Sender Frontend.

The ESP is programmed to host a server (WiFi network) which any personal device can connect to. This allows the device to communicate to the ESP with a small frontend (Fig. 6) at URL 192.168.4.1. Once “Broadcast Sign” is clicked, the ESP receives the entered information via WiFi.

The ESP then transmits this information to the LoRa module with SPI, pulling CS LOW and communicating with MOSI synced to SCLK. This information is used by the LoRa to then broadcast with CSS to the receiver module subsystem.

4.2 Software

Programming is done on the sender and receiver ESP32 chips using an ESP32 auto-downloader (USB-C) connected to a laptop with Arduino IDE v2 (C++). It automatically pulls BOOT low, triggers EN, communicates with TX and receives with RX when prompted by the laptop. Jumper header pins are used to connect the auto-downloader to the chips on the PCBs.

4.2.1 Receiver

The first implementation focuses on reading the motion sensor pin and updating the display accordingly.

For the setup, we initialise variables including the display backlight pin (ILI_LED), motion sensor input pin (PIR) and LoRa interrupt pin (LORA_DIO0)

State 0 is the asleep state - the display is OFF and the ESP goes to sleep waiting on PIR and LORA_DIO0 to wake it up. So it will wake up if motion is detected (and promptly turn on the display) or if a packet is received (and promptly update the display). It will also keep LoRa RES high to prevent resetting the LoRa when asleep.

State 1 is the active high state - ESP ON, the display is ON and motion is still HIGH. Nothing is changing.

State 2 is the sleep timer state - The moment the motion sensor goes back LOW whilst in state 1, it starts a sleep timer (saves the fallTime) and waits until 10 seconds have elapsed. If 10 seconds have elapsed without PIR going HIGH and returning to state 1, the system will go to state 0.

The setup code for the motion sensor is as follows:

```
#define PIR 34
#define ILI_LED 21
const int SLEEPTIME = 10000;
int state = 2;
// state 0 = Asleep (OFF), state 1 = Active High (ON and PIR ON), state
2 = Sleep Timer (ON but about to sleep after SLEEPTIME)
unsigned long fallTime = 0;
```

In the loop, we check the motion sensor pin and change state accordingly.

```
// update motion sensor reading
int motion = digitalRead(PIR);

if (motion == HIGH && state != 1) {
    digitalWrite(ILI_LED, HIGH);
    state = 1;
}

if (motion == LOW && state == 1) {
    fallTime = millis();
    state = 2;
}

if (state == 2 && (millis() - fallTime >= SLEEPTIME)) {
    // keep LoRa receiving
    LoRa.receive();
}
```

```

digitalWrite(ILI_LED, LOW);
delay(100);

// wakeup with motion sensor or LoRa
esp_sleep_enable_ext1_wakeup(WAKEUP_BITMASK,
ESP_EXT1_WAKEUP_ANY_HIGH);

// keep LoRa on (don't reset)
gpio_deep_sleep_hold_en();
rtc_gpio_hold_en((gpio_num_t)LORA_RST);

// sleep
esp_deep_sleep_start();
}

```

To receive packets from LoRa, it parses the packet (first determining if anything was sent by checking the packet size). It then reads the byte stream. The stream is broken up into chunks by a struct, representing the fire danger rating, fire ban status, trail status, fire assembly point direction and distance respectively.

The packet is set up as a struct to easily extract the necessary chunks of information.

```

struct BushfirePacket {
    uint8_t fireRating;
    uint8_t trailClosed;
    uint8_t fireBan;
    uint16_t direction;
    uint16_t distance;
};

BushfirePacket packet;

```

The byte stream is extracted into the related variables.

```

// LoRa receive
int packetSize = LoRa.parsePacket();

if (packetSize == sizeof(BushfirePacket)) {
    LoRa.readBytes((uint8_t *)&packet, sizeof(packet));

    rating = (FireRating)packet.fireRating;
    trailClosed = packet.trailClosed;
    fireBan = packet.fireBan;
    direction = packet.direction;
    distance = packet.distance;
}

```

```

drawScreen();

// also resets sleep timer
fallTime = millis();
}

```

Each loop, the ESP checks if any bushfire safety information has changed since the last loop. If anything has, it will update the screen by calling `drawScreen()`, which wipes the previous screen with a black fill and writes the new display on top.

The rest of the code consists of standard algorithms or helper functions for drawing on the display. The full receiver code is in Appendix [11.1](#).

4.2.2 Sender

The sender creates a web server using the WiFi and WebServer libraries, and uses a HTTP route to receive updates in data.

```

server.on("/", HTTP_GET, [] ()
{
    Serial.println("Browser connected");

    server.send_P(
        200,
        "text/html",
        webpage
    );
});

// Update Route
server.on("/update", HTTP_GET, [] ()
{
    Serial.println("Update received");

    if (!server.hasArg("rating") ||
        !server.hasArg("trail") ||
        !server.hasArg("ban") ||
        !server.hasArg("direction") ||
        !server.hasArg("distance"))
    {
        server.send(
            400,
            "text/plain",
            "Missing parameters"
        );
    }
}

```

```

        return;
    }

    packet.fireRating = server.arg("rating").toInt();
    packet.trailClosed = server.arg("trail").toInt();
    packet.fireBan = server.arg("ban").toInt();
    packet.direction = server.arg("direction").toInt();
    packet.distance = server.arg("distance").toInt();

```

Once these updates are parsed and checked for errors, the packet is sent via SPI to LoRa (SPI and LoRa libraries) and broadcasted.

```

    LoRa.write(
        (uint8_t*)&packet,
        sizeof(packet)
    );

    int result = LoRa.endPacket();

    if(result == 1)
    {
        Serial.println("LoRa TX success");
    }
    else
    {
        Serial.println("LoRa TX failed");
    }
}

Serial.println("Packet sent");
server.send(
    200,
    "text/plain",
    "Broadcast successful"
);
});

```

The full sender code is in Appendix [11.2](#).

4.3 Decisions & Trade-Offs

4.3.1 2S Instead of 1S

With each component of the receiver PCB using 3.3V power, it might seem unnecessary to use 2 batteries in series for 7.4V instead of just one battery at 3.7V. Some benefits of using 2S (or not using 1S) include:

- Optional 5V power rail: Since this was an early decision during development, there was a possibility that components may need to change if any issues arose. For the case where a new component would require 5V, this 5V rail would prove beneficial, and is good to keep as backup as 5V is a very common operating voltage.
- LDO supply voltage: Using 1S would mean having a nominal voltage of 3.7V, and a fully discharged voltage of 3V. Even using a low-dropout device, any voltage below $3.3V + 125mV = 3.425V$ (TLV73333PQDBVRQ1 drop-out voltage = 125mV) would lead to dysfunction of the entire 3.3V rail, instantly turning off or possibly causing damage to the receiver components. The battery will be guaranteed to discharge into drop-out range.

The trade-off for using 2S is the increased BMS and battery holder size. The 2S BMS is 47mm x 23mm, whilst the equivalent 1S BMS is 37mm x 7mm, less than one third the board size. The battery holder is also doubled in size to fit another 18650 battery. This makes the product as a whole larger, requiring a larger enclosure, however this trade-off is taken as it is effectively vital for safe operation.

4.3.2 Buck + LDO Instead of LDO Only

Dropping from 7.4V to 3.3V in the receiver power subsystem can be done via a single LDO, however it was opted to use a two-stage buck + LDO setup instead, where the buck drops to 5V (for possibly useful 5V rail, and well above LDO drop-out voltage), and the LDO drops to a clean 3.3V.

Note that the LDO is considered vital and non-optional as the MCU and LoRa module are especially sensitive with power thus a clean, non-noisy 3.3V regulator is needed.

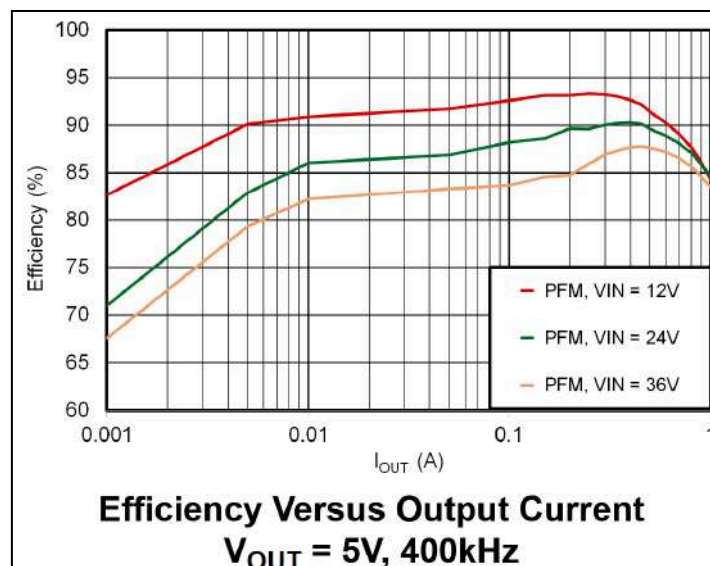


Fig. 7. LMR51610XFQDBVRQ1 Efficiency Versus Output Current @ 5V.

The benefit of using buck + LDO (or not using LDO only) is efficiency, giving the product a longer battery life and reducing the risk of draining before it can recharge. At 100mA (approximate current when display is ON):

- The components draw 100mA at 3.3V, so $P_{out} = 0.33W$.
- The buck converter has an efficiency of $\sim 94\%$ (extrapolating 5V above 12V line in Fig. 7).
- The LDO when used for 5V to 3.3V has an efficiency $\leq \frac{0.1A \cdot 3.3V}{0.1A \cdot 5V} = 66\%$ (maximum efficiency ignoring quiescent)
- The LDO when used for 7.4V to 3.3V has an efficiency $\leq \frac{0.1A \cdot 3.3V}{0.1A \cdot 7.4V} = 45\%$
- Thus, with buck + LDO we have $0.94 \cdot 0.66 = 62\%$ efficiency, so
 $P_{in} = 0.33 / 0.62 = 0.532W$, and thus $I_{in} = 0.532 / 7.4 = 72mA$
 - At 3000mAh battery capacity and 72mA, we have 41.7hrs of runtime
- With just LDO we have constant current 100mA through input and output
 - At 3000mAh battery capacity and 100mA, we have 30hrs of runtime

Thus, we get an extra 11.7hrs of display ON runtime with a two-stage setup, which is the equivalent of multiple days or possibly weeks of extra runtime assuming the motion sensor is not always tripped.

The trade-off with the buck + LDO setup is the use of an extra component, adding about \$3 extra cost for a high-quality buck + passives, and extra PCB component space. The board size however remains under 100mm x 100mm, thus costing the same as it would without the buck in JLCPCB pricing. Thus, the buck + LDO setup was considered to be worth the trade-off.

4.3.3 TFT LCD Instead of E-Paper

A few considerations had to be made when choosing a display that would both satisfy user requirements and benefit their safety, whilst also remaining low-power and easy to implement. Some benefits of using a TFT LCD (or not using E-Paper) include:

- Colour display: E-Paper does not offer a colour display, which makes it much harder for users to get information at a glance. This violates the requirement [2.4.10](#) (Display Attention).
- Backlight: TFT LCDs offer a backlight in order to satisfy requirement [2.4.11](#) (Display Visibility). E-Paper displays do not include this.

The trade-off with using TFT LCDs is the lower efficiency of them, or more notably the high efficiency of E-Paper displays. E-Paper displays preserve their screens without power, and only draw power on updates, which is highly optimal for the given setup where updates don't happen too frequently, and efficiency is highly valued. However, the requirements listed above need to be met, and E-Paper displays do not meet them. Furthermore, a motion sensor can be used to only activate the display when needed, conserving battery at a small extra cost of $\sim \$3$ and marginally more complexity.

4.3.4 LoRa Instead of WiFi/Cellular/Satellite

LoRa modules or satellites are simply vital for the deployment of the product in remote areas, as WiFi and cellular communications are far too low-range. The trade-off is the added complexity of LoRa, requiring extra components (modules) with SPI communication, a frontend and entire sender PCB. LoRa was chosen over satellite to reduce costs, as it requires no maintenance cost and no fees to use.

4.3.5 ESP32 Over Alternatives

ESP32s boast onboard support for WiFi which is highly necessary for the product. A considered alternative was the STM32 chips, however these alternatives do not have WiFi support and would require an extra module, adding complexity and cost.

Furthermore, the group is more experienced with ESP32s, reducing complexity even more. Importantly, processing speed is not considered important for the application, as SPI communication with two peripherals doesn't require expensive signal processing or other operations. A major trade-off is the power draw being higher for ESP32s, however a motion-activated sleep/wake implementation solves this problem with slightly extra complexity (but overlaps as a solution for display power draw).

4.3.6 Buck & LDO selection

One of the major reasons for the buck and LDO selection is simply experience with the components, and access to spares due to previous work with them. However, the buck was also preferred over the ones proposed from the previously created power supply PCB, as the latter requires expensive and massive capacitors due to the buck's lower quality. The chosen buck can operate with 22uF capacitors instead of 47uF, allowing smaller SMD 0603 or 0805 selections to be made for the capacitors. A small but ultimately worthy trade-off is the \$3 cost of the buck itself, however this is nearly completely outweighed by not requiring expensive through-hole 47uF capacitors.

4.3.7 WiFi for Data Upload

In order to meet user requirement [2.4.12](#) (Ease of Use), operators must be able to transmit bushfire safety data within 60 seconds from their location. A simple implementation of uploading bushfire safety data to the sender device would be physical connection (e.g. USB-C) with a personal device, however this would require traveling with or near the device at all times, and would slow down operational use.

WiFi for data upload requires less interaction with the device and allows operators to send data quicker. It is also the cheapest and simplest alternative to other wireless communications due to the inbuilt WiFi support of ESP32 chips. A trade-off is the added complexity of implementing a small frontend for uploading data to the sender device.

4.4 Subsystem Integration & Prototyping

4.4.1 Individual Components

Major components were received and tested in individual units. Surface mounted components for PCB integration were bought with breakout board equivalents. Individual tests included:

- Solar panel output voltage at different light intensities
 - Output ranged from 0.7V (complete darkness) to ~12 maximum. Average was around 9-10V in normally lit rooms (lab rooms)
- ESPs tested with dev boards
 - Compilation and upload of code
 - Testing with motion sensor (debug output statements in serial monitor, e.g. "Motion: 1" for HIGH input and "Motion: 0" for LOW input)
 - Display testing, working with TFT_eSPI library. Enabling backlight, writing text onto display. SPI functional using SPI library.

- Compilation of code working with LoRa modules using LoRa library. Debug statements on successful broadcast and 'ready to receive' state. SPI function using SPI library.

4.4.2 Integration Testing (Breakout)

Receiver and sender modules were tested together as full units with breakout boards (excluding receiver power subsystem).

- Personal device connected to sender ESPs WiFi and sending data.
- Sender ESP connected to LoRa (sender) and successfully transmitted.
- Receiver ESP connected to LoRa (receiver), display and motion sensor.
 - Display successfully turning on and off based on motion activated sleep timer.
 - LoRa successfully receives transmissions from the sender.

Issues encountered were mainly related to LoRa transmissions. Code compilation for sender PCB failed for a while due to mismatched wires between ESP board and LoRa board. Bytes originally came in as gibberish on serial monitors due to baud rate mismatch.

4.4.3 PCB Implementation

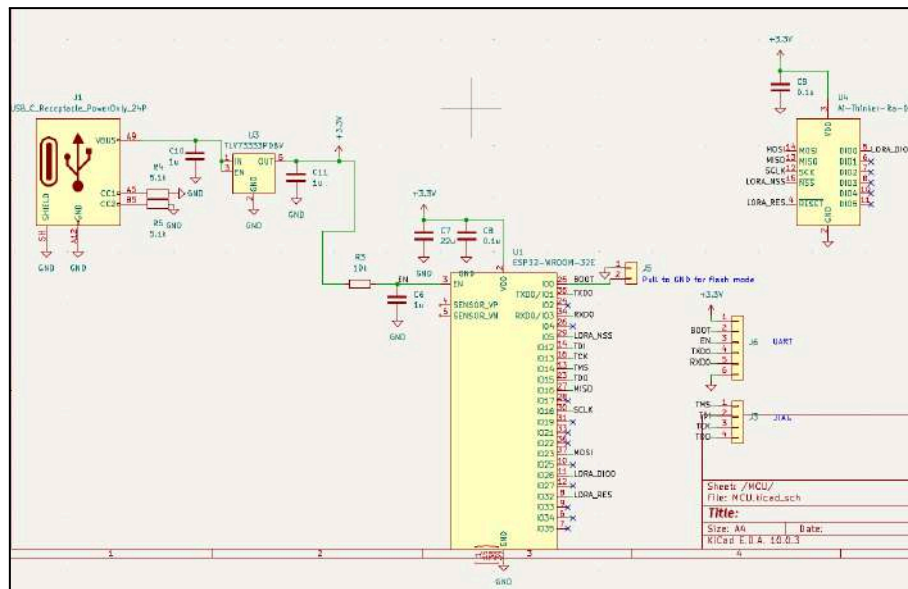


Fig. 8. Sender Schematic.

The sender schematic (Fig. 8) mainly involves the connection of the USB receptacle to the LoRa module and ESP after and LDO to 3.3V. Net labels are used to connect the SPI interface between the ESP and LoRa module. Jumper header pins are used to allow UART communication (auto-downloader) and JTAG if necessary. Proper decoupling is used on the modules, RC is properly used on EN and the USB-C pins are pulled down with 5.1k resistors to indicate a power receiving device.

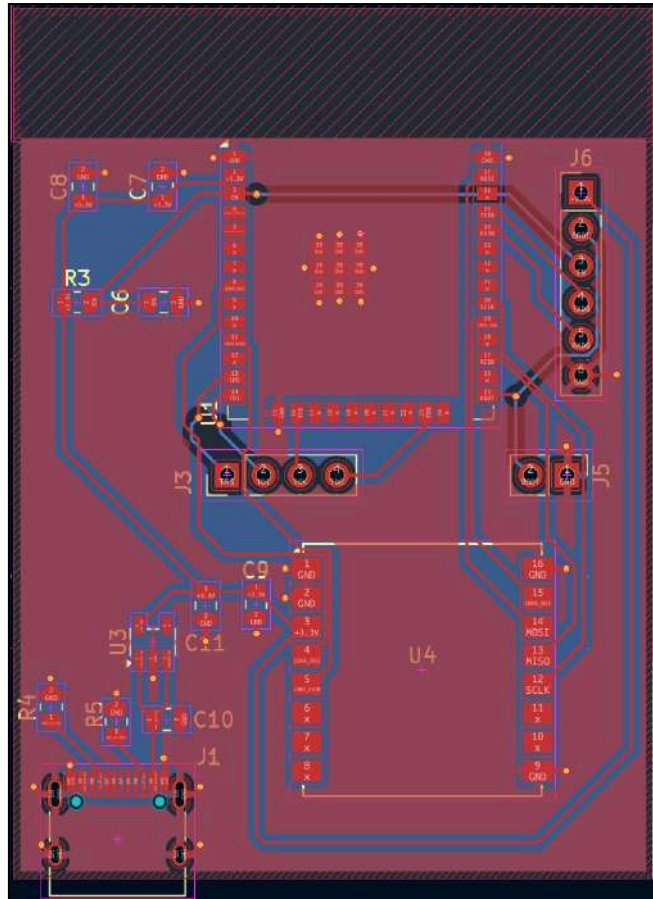


Fig. 9. Sender PCB.

The sender PCB (Fig. 9) is routed with the USB-C receptacle partially off the edge cut to allow connection without the PCB getting in the way. The PCB is 2-layer with ground poured on both layers for low noise and thermal relief. The UART and JTAG connectors are placed close to the ESP for trace simplicity.

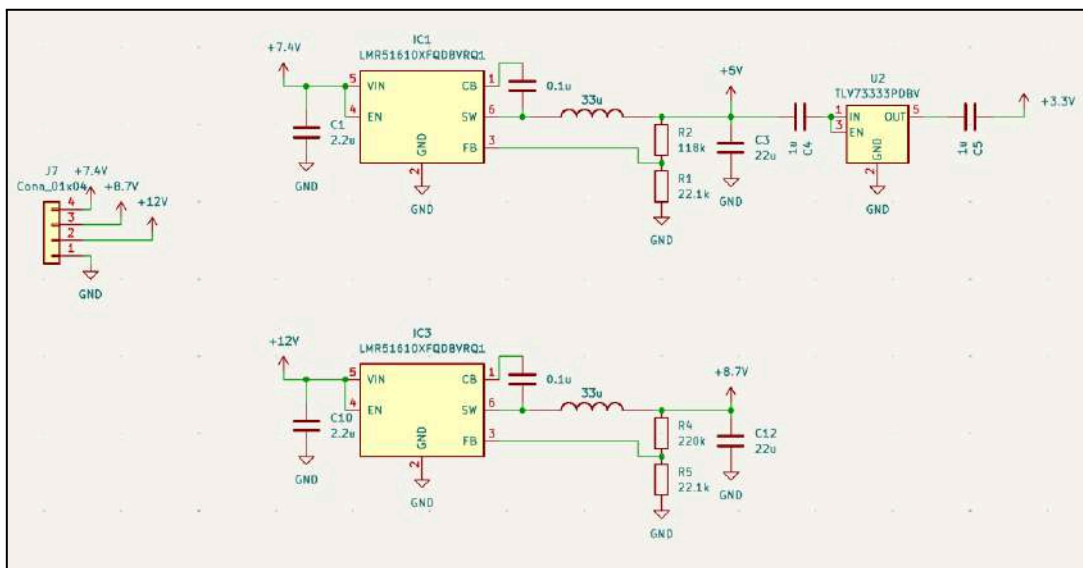


Fig.10. Receiver Power Schematic.

The receiver power schematic (Fig. 10) involves the 12V-8.7V buck converter in the bottom left, with a 220k and 22.1k resistor setup to get the necessary 8.7V. For the given buck, $V_{REF} = 0.8V$, thus:

$$R_{FBT} = \frac{V_{OUT} - V_{REF}}{V_{REF}} \times R_{FBB}, \text{ where } R_{FBB} = 22.1k\Omega \text{ (standard resistor value)}$$

$$R_{FBT} \approx 220k\Omega \text{ (nearest standard value).}$$

Similarly, for the 7.4V-5V buck converter, we get (using $R_{FBB} = 22.1k\Omega$ again):

$$R_{FBT} \approx 118k\Omega \text{ (nearest standard value).}$$

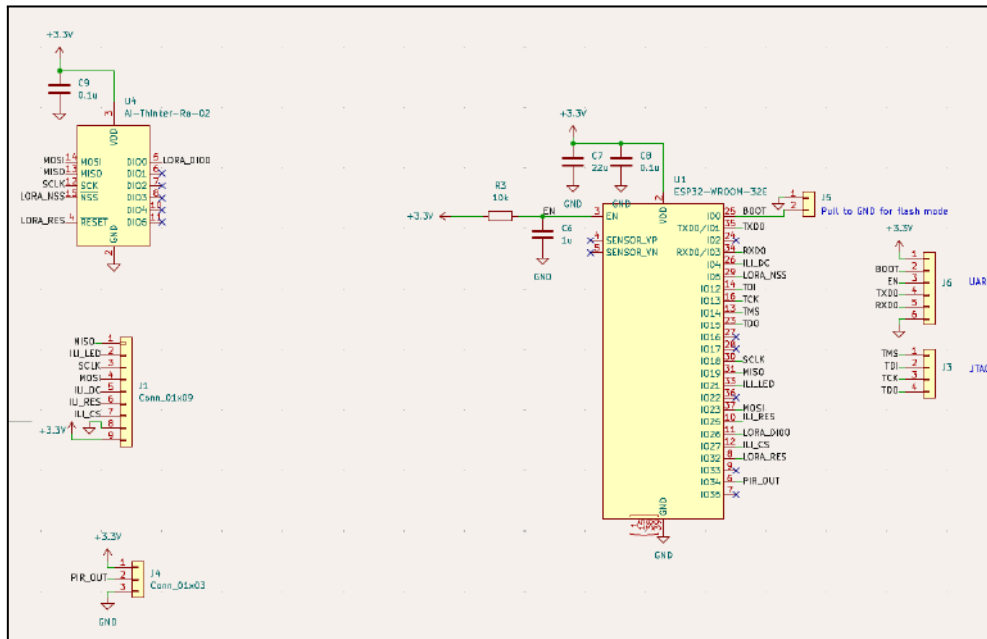


Fig.11. Receiver Module Schematic.

The receiver module schematic (Fig. 11), similarly to the sender schematic, includes an identical setup with the ESP and LoRa, with the addition of header pins for the display and motion sensor, using net labels to connect them to the ESP.

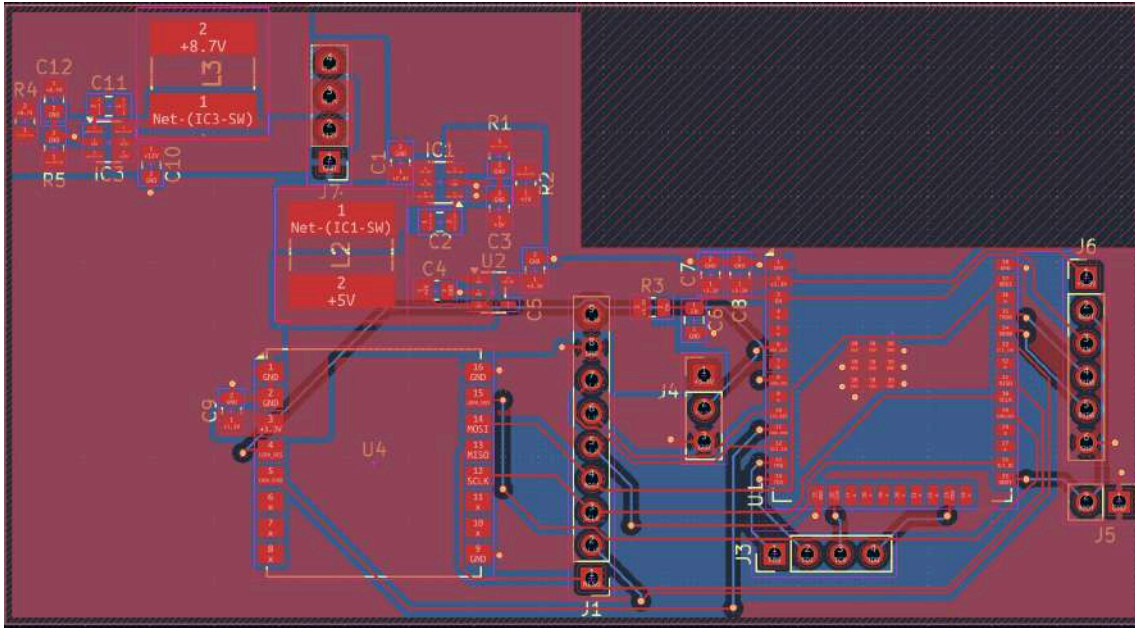


Fig. 12. Receiver PCB.

The receiver PCB (Fig. 12) has schematic placement prioritising simple power connections, then simple/short SPI connections. The top left works with the 12V-8.7V buck, as well as the two-stage 7.4V-3.3V components, using pours whenever possible for thermal relief and noise prevention. Specifically, the buck layouts are applications of the datasheet's recommended layout in Fig. 13.

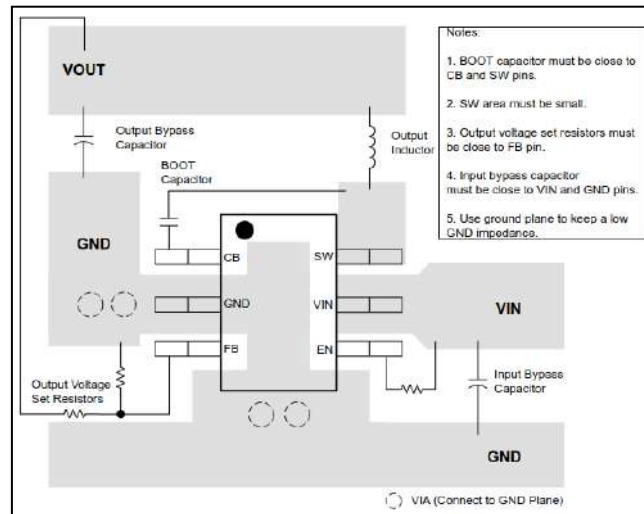


Fig. 13. Buck Converter Layout

The 3.3V pour connects to the LoRa, display, motion sensor and ESP without any discontinuities. The SPI lines travel from the ESP to the LoRa and display with similar lengths to maintain clock sync, notably using no vias.

After integration testing on breakout boards, PCBs were received and soldered. Each PCB was initially tested for power supply functionality and proper buck + LDO functionality.

- Sender working with clean 3.3V.

- Receiver correctly dropping 12V to 8.7V and 7.4V into 5V and clean 3.3V.
- Auto-downloader used to upload code to ESPs. LoRa, motion sensor and display functionality matches that of breakout boards.

A challenge that came up was uploading with the auto-downloader. This was fixed by swapping the TX and RX cables from the downloader board, as TX and RX swap when connecting to the external device.

The Bill of Materials for both PCBs are in Appendix [11.3](#).

4.4.4 Enclosure Installation



Fig. 14. Enclosure.

The enclosure (Fig. 14) aims to provide an IP43 rating for the product, shielding it from damage due to rain, wind or other weather. The motion sensor sits at the top of the enclosure and faces forward. The solar panel sits on a pivot mechanism to angle accordingly. An acrylic cover was installed in the front and back for display visibility and easy showcase of the components inside, however the deployed models would have no back acrylic.



Fig. 15. Final Installation.

Lastly, with the sender and receiver fully functional, the enclosure was printed and all components were placed inside (Fig. 15). The solar panel ground was spliced into a jumper receptacle (for PCB) and wire soldered onto P-, and the solar panel power was soldered onto P+. All other BMS wires from the battery holder were soldered on and wrapped in electrical tape to prevent dangerous shorts involving the battery.

All operations were tested inside the enclosure and successful.

5. Verification Results

5.1 Verification Overview

Verification of the Remote Information E-Display was performed throughout the development and assembly of the prototype. Due to the complexity of implementing the main components onto the custom PCB, verifying the entire system was difficult as a single unit from the beginning of development.

Instead, functionality was verified as the system was developed. Two ESP32 development boards were used during the software and communication development stage. The development-board was useful for verifying the software since the two ESP32s can be configured as the transmitting and receiving devices. This allowed information to be transmitted and the information to be displayed.

The complete system was then integrated with the custom PCB. Electrical verification was performed using a multimeter to probe important points on the PCB and confirm that the required voltages were correct. This was important for the power subsystem because multiple voltage-conversion stages were used.

Therefore, verification of the final prototype consisted of:

- Testing using two ESP32 development boards;
- Testing of the LoRa communication system;
- Display and PIR functionality testing;
- Electrical testing of the custom PCB using a multimeter;
- Verification of PCB design using KiCad DRC/ERC and Gerber inspection;
- firmware flashing onto the completed PCB;
- power-consumption measurements
- runtime comparison between alternative power architectures.

5.2 PCB Design and Electrical Verification

The two ESP32 development boards were used to verify the software and communication components before the complete PCB implementation.

One ESP32 was configured as the transmitting device and the other as the receiving device. The development-board setup was used to verify that:

1. The ESP32 could execute the firmware
2. information could be generated at the transmitting device
3. the information could be transmitted using LoRa;
4. the ESP32 can receive the information;
5. the received information is correct
6. the information could be displayed on the TFT LCD.

This testing was important because the final system relies on the LoRa modules to work. The chain of the system includes:

Information source (web server) → ESP32 → LoRa transmitter → LoRa receiver → ESP32 → TFT display

Testing this chain using development boards allowed software and communication issues to be resolved before attempting to operate the same functionality on the custom PCB.

5.3 Custom PCB Verification

Before the PCB was manufactured, the design was checked in KiCad. The schematic and PCB layout were reviewed using the available electrical and design-rule checking tools and met the assembly capabilities of JLC PCB. The Gerber files were also inspected using the KiCad Gerber Viewer to ensure that the copper layers, board outline and drill locations were correctly generated.

Once the PCB was manufactured, the board was checked before being used as part of the complete system.

A multimeter was used to probe important locations on the PCB. In particular, the power input and regulated power rails were checked to ensure that the voltages were as expected. Connecting an incorrect converter directly to the ESP32, LoRa module or display could damage the components. The voltage outputs were therefore checked before the system was connected.

Outlined in table IV is the table for PCB Verification steps:

TABLE IV

PCB VERIFICATION STEPS

Verification	Method	Result
Schematic verification	KiCad electrical checks	Successful
PCB design verification	KiCad DRC/ERC	No issues preventing manufacture
Gerber verification	KiCad Gerber Viewer	PCB layers and drill locations correctly generated
PCB continuity	Multimeter	No unintended connections found
Power-rail verification	Multimeter probing	Voltage rails generated
Firmware programming	ESP32 programming interface	Firmware successfully implemented
Integrated operation	Connected PCB without major components	System functionality demonstrated

5.4 Power Subsystem Verification

The power subsystem was a critical part of the prototype because the Remote Information E-Display is required to operate without reliance on grid power. The system was therefore designed around a solar-charged 2S lithium-ion battery.

The battery has a nominal voltage of 7.4 V, with the measured battery voltage during testing reaching 8.06 V. The higher measured voltage is consistent with a 2S lithium-ion battery being charged.

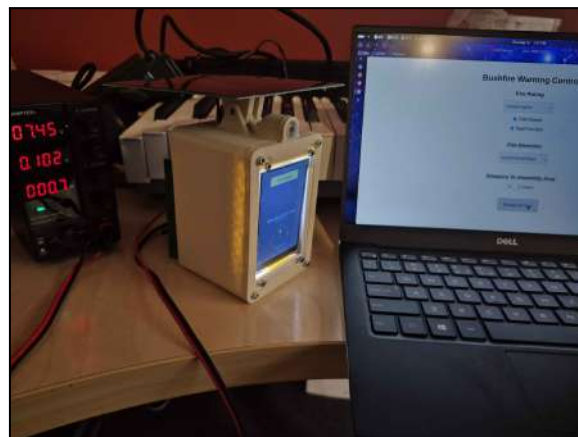


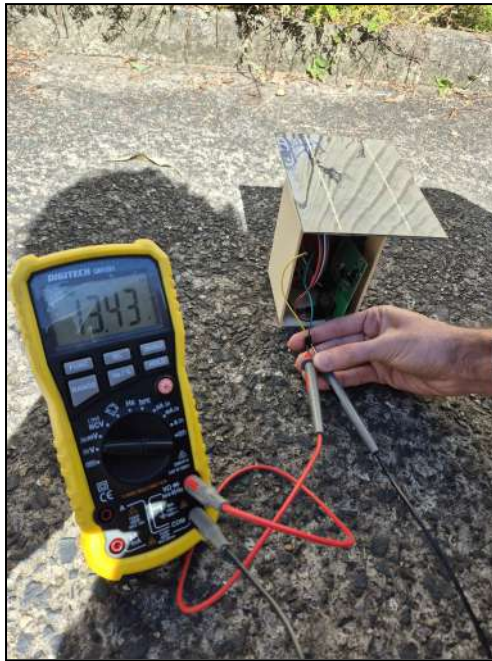
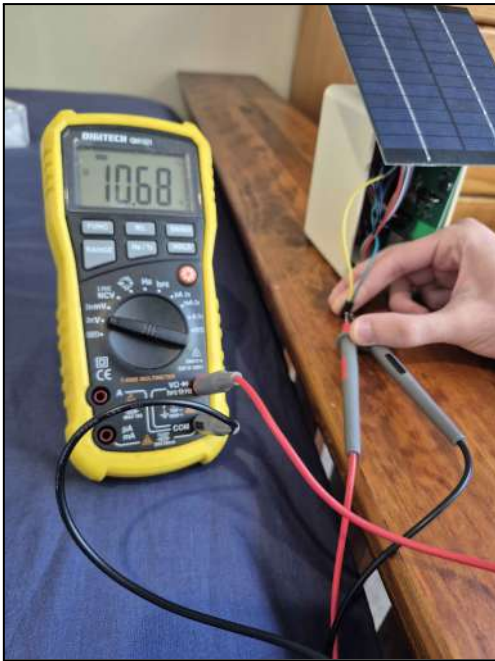
Fig. 16. Testing Setup.

The solar panel was also measured under different lighting conditions to assess the voltage available to the charging system.

TABLE V

SOLAR PANEL & BATTERY MEASUREMENTS

Test Condition	Measured Voltage
Solar panel - well lit	10.68V
Solar panel - fully lit	13.43V
Battery	8.06V
Battery nominal voltage	7.4V



These measurements demonstrate that the solar panel is capable of producing a voltage above the nominal battery voltage under suitable lighting conditions. These measurements provide evidence that the solar panel and battery subsystem are capable of operating at the voltage levels required by the prototype. However, they do not establish the long term charging performance of the system under changing weather conditions.

5.5 Power Conversion Verification

The final architecture uses a buck converter followed by an LDO. The buck converter is used to reduce the battery voltage, whilst the LDO provides a 3.3V supply for the ESP32 & LoRa SPI communication.

The power design was initially calculated during the power-design stage. The buck converter was designed for an approximately 5 V output, with a minimum calculated inductance of 70.5 μH and minimum output capacitance of approximately 30.9 μF . A 220 μH inductor and 470 μF output capacitor were implemented.

TABLE VI

POWER CONVERSION

Parameter	Calculated value	Implemented Value
Switching frequency	-	40.5kHz
Peak switch current	1A	Below 1.5A which is the device maximum
Current sense resistance	0.3 Ω	0.33 Ω
Minimum inductance	70.5 μH	220 μH
Minimum Output Capacitance	30.9 μF	470 μF
Output voltage	5V	~5.21V

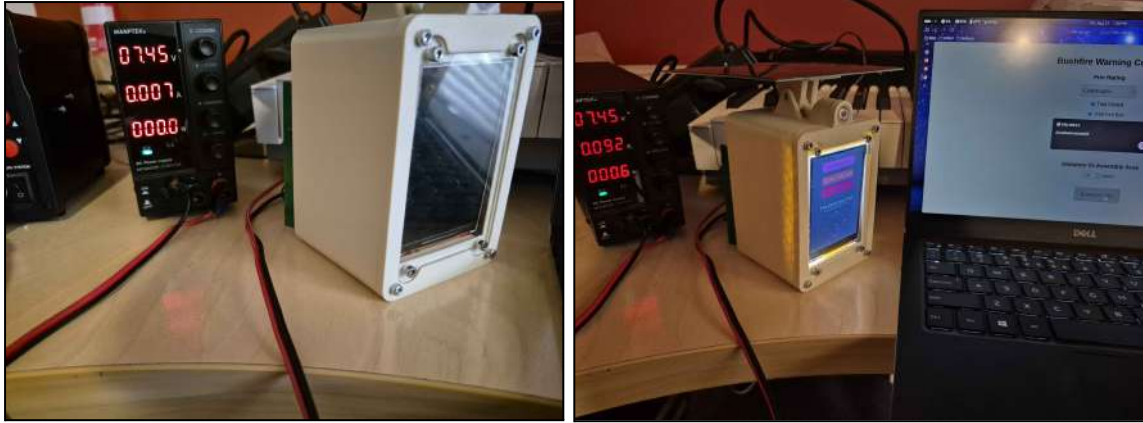
5.6 Power Consumption

Power consumption was measured with the system operating in both active and sleep conditions. This was important because the product is intended to operate autonomously from solar energy and battery storage.

TABLE VII

POWER CONSUMPTION

Operating Conditions	Measured Current	Measured Power
Display Active	91mA	0.6W
Sleep Mode	7mA	0W



The sleep current shows a significant reduction compared with the active current:

$$\text{Current reduction} = \frac{91-7}{91} \times 100 = 92.3 \%$$

Therefore, the system reduced its measure consumption when transitioning from active to sleep mode. This result supports the decision to implement motion based power management.

5.7 Runtime Verification

The power architecture was also assessed by comparing the runtime of the LDO with the final buck plus LDO architecture

TABLE VIII

RUNTIME VERIFICATION

Power Architecture	Measure Runtime
LDO only	10 hours
Buck + LDO	13.75 hours
Increase	3.75 hours
Percentage Increase	37.5%

Although the buck converter introduces additional components, the increased runtime provides a benefit for an autonomous system.

5.8 LoRa Communication Verification

LoRa communication was verified using two ESP32 development boards before the system was integrated into a custom PCB. During testing transmitted packets were received rapidly and no packet loss was found. The typical RSSI observed was about 65. The successful communication demonstrates that LoRa was capable of transmitting small information packets.

For future verification, a fixed number of packets should be transmitted at increasing distances, with packet loss and RSSI recorded at each distance.

5.9 Automated Information Update Verification

The purpose of the functionality is to allow information to be updated remotely without requiring a ranger. The information transmitted by the prototype includes:

- Fire danger rating
- Fire ban status
- Trail status
- Direction
- Distance to the relevant assembly point

This is an important distinction from traditional manually updated signs. The display does not require physical access whenever the information changes.

5.10 Verification Against Technical Design Requirements

The results can be compared against the requirements established during the Technical Design Review.

TABLE IX
VERIFICATION AGAINST TDR

Technical Design Requirement	Verification Evidence	Result
No reliance on grid power	2S battery operation and solar panel measurements	Verified in the prototype level
No direct internet requirement at display	LoRa communication between transmitter and display	Verified
Low power operation	91mA active, 7mA sleep	Verified
Solar Charging	10.68V - 13.43V measured from the solar panel	Partially Verified
Automated Updates	LoRa transmission and display update	Verified
Remote Communication	Successful LoRa communication	Verified under test conditions
Symbology	Symbols implemented and Displayed	Functionally verified
Motion based activation	PIR testing	Verified
Weatherproof enclosure	3D-printed enclosure	Not really verified

	implemented	experimentally
Long term autonomous operation	No extended field test	Not verified

The majority of the core functional requirements were shown by the prototype. The main requirements that remain unverified are those requiring field testing.

5.11 Test Limitations

Although the prototype demonstrated the core system functionality, several limitations affected the verification process.

The first limitation was the difficulty of testing the complete system independently from the beginning. This means that much of the system verification was performed progressively as the PCB was assembled.

The use of two ESP32 development boards helped address this limitation to be tested independently of the final PCB.

A second limitation is that the LoRa testing didn't establish the maximum communication range. The measured RSSI demonstrates successful communication under the tested conditions but cannot be used alone to guarantee reliable communication especially in remote deployment

The solar measurements were also taken under specific lighting conditions rather than continuously throughout a complete day. Therefore, the results show that the solar panel can generate a voltage above the battery voltage under the tested conditions

Finally, weatherproofing was also not tested. Although a 3D-printed enclosure was used to house the electronics, no outdoor testing was performed.

6. Standards & IP

6.1 Standards & Safety

6.1.1 LoRa Radio Communication

Firstly, the LoRa device must operate within the boundaries proposed by the Radiocommunications (Low Interference Potential Devices) Class Licence 2025, under section 132 of the *Radiocommunications Act 1992*.

19	Any	433.05–434.79 MHz	25 mW	None
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Fig. 16. Table 1 of Radiocommunications (Low Interference Potential Devices) Class Licence 2025.

From the operating parameters, only certain frequencies are permissible for all classes of radiocommunications transmitters. The LoRa module broadcasts at 434MHz, keeping within the bounds of item 19 as per Fig. 16.

RF_OP	RF output power in 50 ohms on RFO pin (High efficiency PA).	Programmable with steps	Max	-	+14	-	dBm
			Min	-	-1	-	

Fig. 17. SX1278 Output Power on RFO pin.

The license also restricts the maximum EIRP (Effective Isotropic Radiated Power) to 25mW. The SX1278 LoRa module uses a maximum output power of 14dBm through the RF_OP (high efficiency, Fig. 17) pin which is used in this application (by setting the RFO pin to 1 and PA_BOOST to 0 on LoRa.h in the LoRa library).

This RFO pin provides an equivalent maximum of ~25mW of output power, staying within the set boundaries. However, if more output power is desired for better range then higher frequencies may be used due to higher maximum EIRPs.

6.1.2 Enclosure IP Rating

Ingress protection (IP) ratings guide

IP ratings are represented by combining the first and second digits of the below columns

1 st numeral - solid foreign objects		2 nd numeral - water			
0	No protection		0	No protection	
1	Protected against solid foreign objects of 50 mm Ø and greater		1	Protected against vertically falling water drops	
2	Protected against solid foreign objects of 12.5 mm Ø and greater		2	Protected against vertically falling water drops when enclosure tilted up to 15°	
3	Protected against solid foreign objects of 2.5 mm Ø and greater		3	Protected against spraying water	
4	Protected against solid foreign objects of 1.0 mm Ø and greater		4	Protected against splashing water	
5	Dust-protected		5	Protected against water jets	
6	Dust-tight		6	Protected against powerful water jets	
Example:  + 		7		Protected against the effects of temporary immersion in water	
IP 65 → Protected against water jets		8		Protected against the effects of continuous immersion in water	
→ Dust-tight		9		Protected against high pressure and temperature water jets	

IEC Making electrotechnology work for you.

Fig. 18. IP Rating Guide (IEC).

Under IEC 60529 (Fig. 18), an IP rating of IP43 requires the enclosure to be protected from solids of $\geq 1\text{mm}$, and water from an angle less than 60 degrees from vertical (sprayed).

6.1.3 Li-Ion Battery Testing

The following test procedures are included in UN 38.3:

- T1 – Altitude Simulation (Simulation to 50,000 ft altitude atmosphere)
- T2 – Thermal Test (12-hour dwell times at -40°C and 72°C, repeated 10 times)
- T3 – Vibration (1G from 7Hz to 18Hz, 2g from >18Hz to 200Hz, 3-hour test in each axis)
- T4 – Shock (34.6g shock pulses in positive and negative direction)
- T5 – External Short Circuit (Pack conditioned at 57°C, short maintained for 1-hour or more)
- T6 – Impact (9.1kg mass dropped from 61cm)
- T7 – Overcharge (Charge at twice manufacturer's recommended current)
- T8 – Forced Discharge (12 V power supply connected in series at maximum recommended current)

Fig. 19. UN 38.3 Li-Ion Testing.

Under UN 38.3 (Fig. 19), the lithium batteries must be tested before usage. Since many of these tests are infeasible mainly due to requiring expensive equipment, our batteries must be sourced from manufacturers that follow these guidelines.

6.1.4 SPI and Buck Switching (EMI)

Parameter	Class A	Class B	Frequency Range
Conducted emissions (Quasi-Peak, QP)	79–73 dB μ V	66–56 dB μ V	150 kHz – 30 MHz
Radiated emissions @ 10 m	40 dB μ V/m	30 dB μ V/m	30 – 230 MHz
Radiated emissions @ 10 m	47 dB μ V/m	37 dB μ V/m	230 – 1,000 MHz
Radiated emissions >1 GHz (3 m)	70 dB μ V/m (peak), 50 dB μ V/m (average)	70 dB μ V/m (peak), 50 dB μ V/m (average)	1 – 6 GHz (typical)

Fig. 20. EN 55032 Requirements.

Under EN 55032 (Fig, 20), the buck and SPI lines (components/traces at high frequencies) must follow guidelines for Class A devices, since hiking trails are not close enough to major residential areas and pose less of a risk for disruption. The buck at 400kHz must be below 79 – 73 dB μ V QP. The SPI lines at ~20MHz must also be below follow the same requirements (below 79 – 73 dB μ V QP).

6.1.5 PCB Manufacturing Tolerances

PCB design must adhere to KiCAD DRC specifications to allow for proper manufacturing under brands such as JLCPCB. These specifications also prevent overheating and electrical failure due to high resistance, thin traces. Pours are prioritised for power stages to minimise heat loss and allow correction for voltage ripple.

6.2 IP Review (Existing Patents)

TABLE X

EXISTING PATENTS

Patent	Inventor	Similarities	Differences
US9646518B2	Troy Raymond Wollard	Implementation of fire danger ratings on an illuminated sign. Use of a solar panel and wireless updates	Uses inbuilt sensing modules, such as temperature, wind speed, wind direction and brightness.
CN103886706A	赵剑毅 梁柱 卢毅	Implementation of fire alerts with a display.	Also uses an alarm, does not display fire danger ratings or bushfire safety information, uses sensors. Similar to a fire alarm.
US20140290103A1	Bradley Robert KYBERD	Displays fire danger rating with a solar panel and wireless updates.	Uses cellular instead of LoRa, uses a needle powered by a motor instead of purely a display.
CN117275173A	王心怡	Use of display for fire danger information.	Use of sensors for extra input, use of cellular. Alerts sent to phones.

6.3 Patent Comparisons

6.3.1 US9646518B2 (Roadside LED Fire Danger Rating Sign)

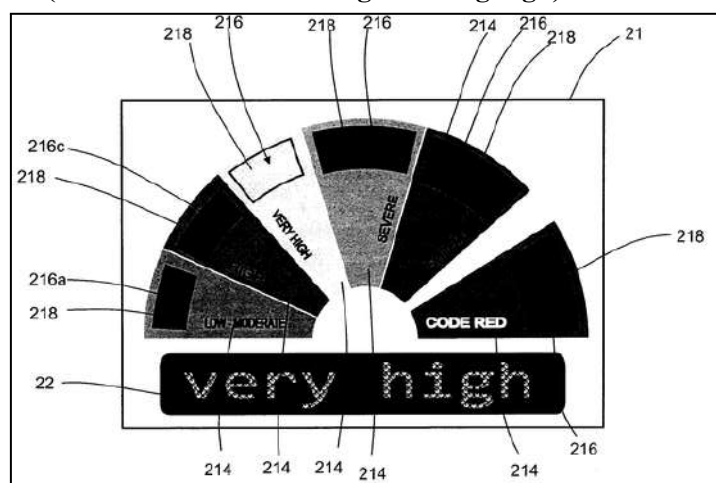


Fig. 21. US9646518B2.

Fire danger rating sign (Fig. 21) that receives data via cellular upload, along with information from internal sensors (temperature, brightness, wind speed and direction). Displays rating with LEDs along with text display.

Possible infringements include the wireless updating and solar battery charging. Using LoRa modules instead of cellular creates a distinction, as well as the BMS architecture. Our product's display is significantly different from the patent's, and includes different and more information. Furthermore, the AFDRS fire danger ratings are distinct from US ratings (e.g. "Very high" does not exist in NSW).

To lower infringement risk, our product should refrain from using LEDs placed on fire ratings, thus continuing to use the display as the full front piece is recommended. Our product should also refrain from using cellular communications or brightness sensors (temperature, wind speed and direction would be considered necessary for further accuracy and would be less likely to infringe).

6.3.2 US20140290103A1 (Motorised Fire Danger Rating Display)

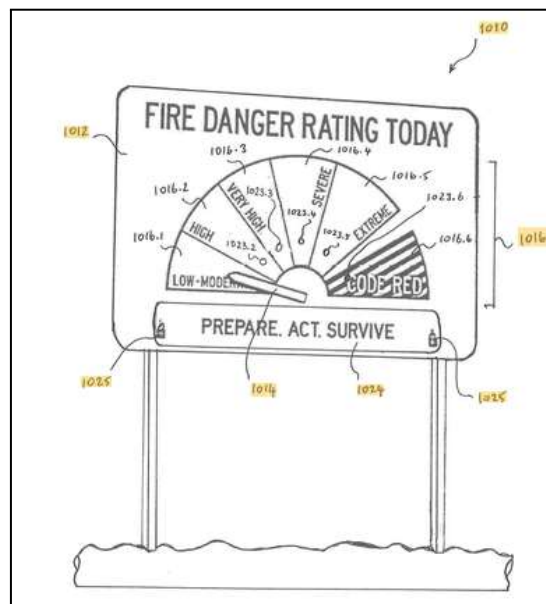


Fig. 22. US20140290103A1.

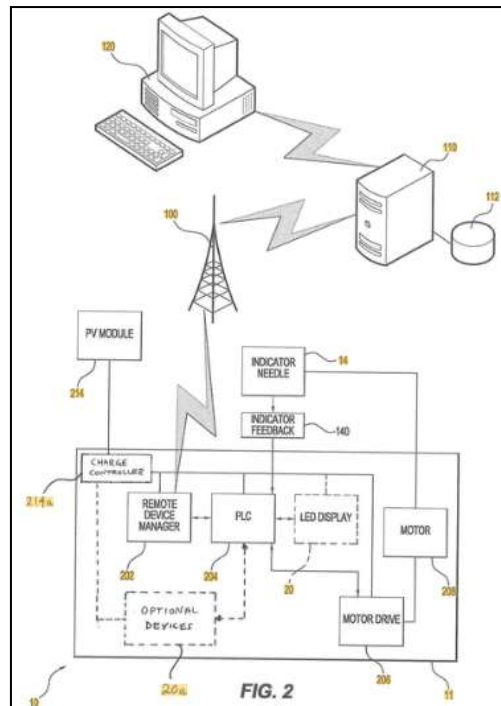


Fig. 23. US20140290103A1 HLD.

Roadside fire danger rating sign (Fig. 22) using a motor to control the needle. Powered by solar and updated purely with cellular communications, our product has many similarities with the product. The patent uses a main host to broadcast information to the displays (Fig. 23). It is also designed to be retrofitted onto existing signs for cheaper and easier deployment.

However, our product completely removes the use of a motor, only using a motion sensor as any peripheral device besides the main operation (display + LoRa). Similarly to other patents, this one also uses cellular which is distinct from LoRa communication to allow for remote deployment. Our product also includes extra bushfire safety information.

To lower infringement, our product should prevent retrofitting deployment onto existing signposts. Our product should also keep focus away from motorised components to prevent infringement on the motorised needle.

6.3.3 CN117275173A (Early Fire Warning System)

No diagrams are provided.

The early warning system consisted of monitoring with smoke detectors, thermal sensors and camera. Activates a sound and display alarm, as well as cellular alerts to all devices nearby. Use of wireless communications for fire information has a slight chance of infringement.

However, our product is quite distinct to this, with a focus on bushfire safety on hiking trails and no use of danger-detecting sensors. Furthermore, our device is solar powered and communicates via LoRa instead of cellular.

To lower infringement, our motion sensor should be kept as PIR and our product should not use a camera or smoke sensor. Furthermore, our device should keep using ESP32 as the patent currently specifies an Arduino Mega2560.

6.4 IP Strategy

6.4.1 Power & Power Saving Architecture

The multi-stage power system involving the 12V-8.7V buck, BMS, 2S Li-Ion battery setup, 7.4V-5V buck and 5V-3.3V LDO architecture should be patented, as well as the motion sensing wake-up system to prevent cheap replications of the product dominating the market. Thus, an Australian provisional patent should be filed followed by a standard patent once the design is finalised.

6.4.2 LoRa Communication

The LoRa communication software and implementation should be kept as a trade secret to prevent malicious use of the LoRa radio frequencies. Without making this trade secret, anyone with a LoRa module could upload data themselves and risk public safety with false information.

6.4.3 Hardware & Enclosure

The enclosure of the product should be made open source under the Creative Commons BY-SA 4.0 license to promote communities creating infrastructure (signposts) that support the correct mounting system and architecture of the product. Accreditation must be made, promoting our product's marketability and influencing more people to apply the correct infrastructure.

7. Business & Development Plan

The next stage of development of the Remote Information E-Display is to transition the prototype into a manufacturable and commercially viable product suitable for deployment across remote public areas. This requires improvements to the electrical, mechanical and software systems, followed by field validation, compliance testing and manufacturing optimisations.

7.1 Current Prototype Limitations & Future Improvements

Although the current prototype demonstrates the core functionality of the Remote Information E-Display, several limitations must be addressed before the commercial deployment of the product. This includes:

7.1.1 Power System

The current prototype demonstrates solar powered operation using a 10W solar panel, lithium ion battery storage and multiple voltage conversion stages. However long term, the operation in outdoor environments has not been demonstrated over seasonal variations in sunlight & temperature.

Future developments should therefore include:

- Optimisation of the solar panel and battery
- Battery state of charge monitoring
- Improved power management
- Battery protection & replacement of equipment procedures
- Long term field testing

The PIR sensor should also be integrated into the power management system more too.

7.1.2 Display

The current TFT display provides the required functionality for the prototype but is not optimal for a commercially outdoor product. To be more commercially viable, the product should have more sunlight readability, lower power and a more durable display.

Potential alternatives should include:

- Transflective LCD technology
- High brightness outdoor displays
- E-Paper technology where updates frequently happens
- Automatic brightness control

The display should be able to sustain long term testing to UV exposure, temperature cycling and moisture from external environments.

Existing commercial fire-danger signs show that solar power, remote updating and environmental protection can be incorporated into outdoor information displays. See this [link](#)

7.1.3 Enclosure & Environmental Protection

The enclosure should be designed to protect the electronics from water and dust and withstand temperature variation and UV exposure.

The enclosure should also provide:

- Secure mounting
- Replaceable battery access
- Cable sealing
- UV resistant material
- Simplified maintenance access

This is important because the design target is for maintenance to be no more than two years.

7.1.4 Communication & Information Management

The current system uses a sender ESP32 and LoRa receiver architecture. Future development should improve the reliability and security of the communication system. Improvements could include:

- Acknowledgement of received information
- Communication fault detection
- Authentication
- Automatic recovery following communication or power failures

7.1.5 Manufacturability

The current prototype demonstrates the feasibility of the product rather than mass production. Future development should have the electronics into a production PCB and simplify assembly.

The production should therefore be:

1. Reduce wiring and connectors;
2. Minimise PCB and enclosure assembly
3. Use commercially available components
4. Design the enclosure for repeatable manufacturing
5. Establish a standardised manufacturing and testing procedure.

7.2 Five-Year Development Roadmap

The five year development plan is shown below:

TABLE XI

FIVE YEAR DEVELOPMENT PLAN

Year	Development Stage	Key Activities	Target Outcome
Year 1	Engineering Refinement	Improve PCB, power system, enclosure, software and display	Engineering prototype
Year 2	Validation	Environmental, power, testing	Completed pre production design
Year 3	Pilot deployment	Deploy units at selected trails / public locations	Field testing of product
Year 4	Manufacturing	Optimising production, supply chain and assembly	Low volume commercial production
Year 5	Commercialisation	Expand deployment and improve product	Product at industrial scale

7.2.1 Year 1 - Engineering Refinement

The first year should focus on the limitations in the current prototype. All the systems should be redesigned for reliability and manufacturability.

7.2.2 Year 2 - Validation

The second year should focus on validating the redesigned system under environmental conditions. The design should not progress to widespread deployment until these tests are met:

- solar charging
- Battery sustainability
- communication reliability
- temperature cycling
- water and dust
- display readability
- long-term operation

7.2.3 Year 3 - Pilot Deployment

A small number of units should be installed at some remote trails or public locations. Data should be collected on:

- Battery performance
- Communication reliability
- Display faults
- Maintenance requirements
- User interaction

7.2.4 Year 4 - Manufacturing

Following pilot testing, PCB assembly, enclosure manufacturing and final system assembly should be outsourced wherever possible.

7.2.5 Year 5 - Commercialisation

The fifth year should focus on expanding deployment and make the product a commercially available remote information system.

7.3 Development & Manufacturing Costs

The following costs are estimates of the expenditure required to commercialise the prototype. The estimates were developed using a bottom-up-cost approach, where the major development activities were identified and assigned estimated costs.

7.3.1 Initial Development Costs

TABLE XII

DEVELOPMENT COSTS

Development	Estimated Cost
Electrical & PCB design	\$30,000
Software & Communication Development	\$20,000
Enclosure redesign	\$20,000
Reliability testing	\$25,000
Manufacturing & Tooling	\$25,000
Field trials and installation	\$15,000
Compliance & certification preparation	\$15,000
Total initial development cost	\$150,000

The \$30,000 electrical and PCB design covers redesign of the prototype electronics for commercial production.

The \$20,000 software budget covers development of updated firmware

The \$20,000 enclosure redesign allowance accounts for the mechanical engineering required to develop a weather-resistant enclosure suitable for long-term outdoor deployment.

The \$25,000 reliability testing covers environmental and operational testing.

A further \$25,000 is allocated to prototype manufacturing and tooling to allow multiple iterations of the PCB, enclosure before production.

The remaining \$15,000 for field trials and installation allows the system to be evaluated at representative deployment locations, while \$15,000 is allocated to compliance and certification preparation.

These values are assumption based allowances and would require quotations from relevant suppliers and testing facilities during commercialisation.

7.3.2 Production Unit Costs

The prototype BOM provides a baseline for the cost of the system. However, the commercial production cost will be significantly higher than the raw component cost because it must also account for PCB fabrication and assembly, mechanical manufacturing, enclosure integration, testing, installation, logistics, warranty provisions and manufacturing overhead.

TABLE XIII

ESTIMATED UNIT COSTS

Production Stage	Estimated Unit Cost
Prototype / very low volume	\$10,000
Pilot Production	\$8,000
50+ units/year	\$7,000
100+ units/year	\$8,000

7.4 Market Position & Selling Price

Existing Australian fire danger signs show that there is an established market for updated, public safety signage. For example, Axent's Australian Fire Danger Rating Sign uses solar power and battery storage. See this [link](#)

The NSW Government has also deployed more than 200 digital fire warning signs that are solar powered and automatically updated. See this [link](#)

7.5 Profitability & NPV

A five year financial model can be used to estimate the commercial potential of the product. The following model assumes that the initial development investment is \$150,000, selling price is \$15,000/unit, production cost reduces from \$9,000 - \$6,000, a 10% discount rate for NPV and no additional financing costs or tax included.

TABLE XIV

COMMERCIAL POTENTIAL

Year	Units Sold	Revenue	Production Cost	Development / Operating Costs	Net Cash Flow
Initial	-	-	-	\$150,000	-\$150,000
1	5	\$75,000	\$45,000	\$30,000	\$0
2	15	\$225,000	\$120,000	\$50,000	\$55,000
3	30	\$450,000	\$210,000	\$80,000	\$160,000
4	60	\$900,000	\$390,000	\$120,000	\$390,000
5	100	\$1,500,000	\$600,000	\$160,000	\$740,000

Using a 10% discount rate, the estimated five-year NPV is approximately \$742,000. The model suggests that the product could be commercially viable if the selling price, production costs and sales volume are achieved. However, this result is dependent on successful field validation and manufacturing cost reduction. It should be noted that these figures should be assumption based rather than a forecast

7.6 Work Breakdown Structure

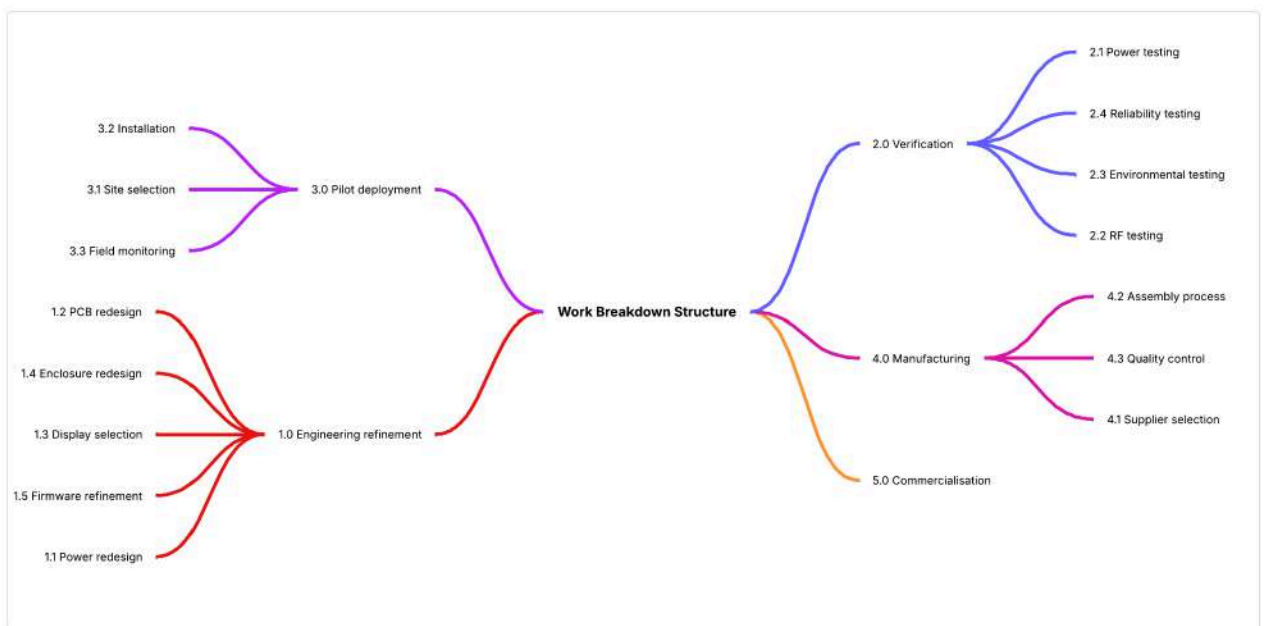


Fig 24: Work Breakdown Structure

The following WBS provides tasks required to take the prototype to market.

Attached below is also the table included for the following Work Breakdown Structure:

TABLE XV

WBS CRITERIA

WBS	Task	Deliverable	Complete Criterion
1.0	Engineering refinement	Production design	All prototype limitations addressed
1.1	Power redesign	Final power subsystem	Required energy budget achieved
1.2	PCB redesign	Production PCB	PCB manufactured and tested
1.3	Display selection	Commercial display	Outdoor readability demonstrated
1.4	Enclosure redesign	Production enclosure	Environmental protection demonstrated
1.5	Firmware refinement	Production firmware	Fault recovery and communication handling implemented
2.0	Verification	Validation report	All defined tests passed
2.1	Power testing	Energy-performance data	Required operating endurance achieved
2.2	RF testing	Communication results	Required communication range achieved
2.3	Environmental testing	Environmental test report	No environmental failures
2.4	Reliability testing	Reliability report	Required long-term operation demonstrated
3.0	Pilot deployment	Field prototypes	Pilot units deployed
3.1	Site selection	Deployment locations	Suitable sites approved
3.2	Installation	Installed units	Units operational
3.3	Field monitoring	Performance data	Minimum monitoring period completed
4.0	Manufacturing	Production process	Repeatable assembly established
4.1	Supplier selection	Approved suppliers	Critical components sourced

4.2	Assembly process	Manufacturing procedure	Production process documented
4.3	Quality control	Quality assurance procedure	Each unit passes production testing
5.0	Commercialisation	Market-ready product	Product ready for sale

7.7 Gantt Chart

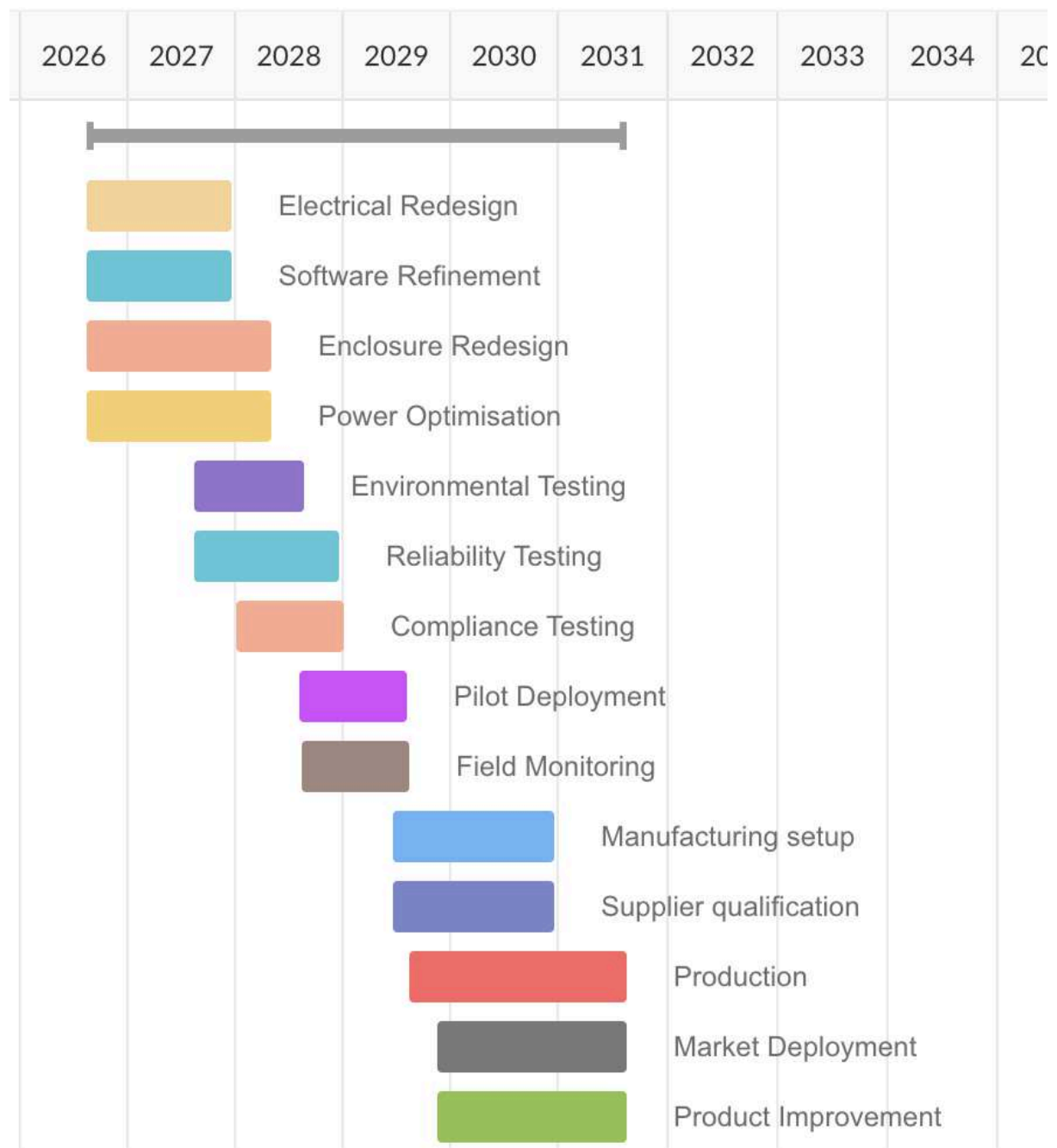


Fig. 25: Gantt Chart 5 Year Development Plan

Here the Gantt chart outlines the proposed five year development and commercialisation plan for the Remote Information E-Display. The initial stage focuses on refining the existing prototype. This includes:

- Electrical Redesign
- Software Refinement
- Reliability and Compliance Testing

The later stages go from pilot to deployment testing, field monitoring, manufacturing setup, supplier qualifications and production. After that, the product then goes to market deployment and product improvements too.

7.8 Additional Resources Required

To go from a prototype to a market ready product, it will require resources that are not available through the prototype alone. This includes, engineering, testing, manufacturing and field deployment resources. These resources are required during the engineering refinement of the five-year development plan.

7.8.1 Electrical Development

TABLE XVI

EXTRA RESOURCES REQUIRED

Resource	Purpose	Proposed Access
PCB Design Software	Redesign schematic and PCB layout for production	Software licence
PCB prototyping service	Revised PCB prototypes	External PCB manufacturer
Oscilloscope	Verify SPI, power rails and communication signals	Electronics Laboratory
Multimeter	Voltage, current and continuity measurements	Electronics Laboratory
DC Power Supplies	Testing of individual subsystems	Electronics Laboratory
RF test equipment	Verify LoRa transmitter/receiver performance	RF laboratory

The existing prototype can be used to develop and debug the initial system.

7.8.2 Power System Development

An autonomous power system is critical to the product because the design must operate without grid power. Therefore some additional resources would need to be included such as:

- Electronic load
- Current and voltage measurement equipment
- Solar panel test equipment
- Temperature sensors

- Data logging equipment
- Battery protection equipment

7.8.3 Enclosure Testing

The current prototype is intended for outdoor use, meaning the enclosure and electronics must be tested against environmental conditions before commercial deployment. The project should outsource formal environmental testing to an accredited laboratory. [NATA-accredited laboratories](#) in Australia provide environmental capabilities, showing us that this type of specialist testing can be obtained externally.

7.8.4 RF and Communication Testing

The current prototype demonstrates the use of LoRa for communication between the sender and receiver. Commercialisation requires more testing to have a more reliable operation in the environment. Some required resources would include:

- RF Spectrum analyser
- Long range antennas
- RF power measurement equipment
- GPS equipment for field range measurements

7.8.5 Display Development

Resources required for the TFT display include:

- Outdoor display samples
- Brightness measurement equipment
- Light metre
- Weathering test facilities

The final display should be selected based on performance rather than keeping the performance of the prototype product.

7.8.6 Manufacturing Resources

Manufacturing should transition from prototype assembly to a production process.

TABLE XVII

MANUFACTURING RESOURCES

Manufacturing Resource	Requirement
PCB Manufacturer	Production of the electronics board
PCB assembly service	Automated placement and soldering
Enclosure manufacturer	Weather resistant enclosure
LoRa/ESP32 supplier	Supply of processing electronics
Battery supplier	Battery packs and protection circuitry from a trusted company
Solar-panel supplier	Production of solar panels

Cable / Connector supplier	Weather resistant wiring and connectors
QA / Testing facility	Testing of every manufactured unit

7.8.7 Compliance & Certification Resources

The Commercialisation process will require access to compliance testing and certification resources.

[The Australian Communications and Media Authority \(ACMA\)](#) states that suppliers must demonstrate compliance, maintain supporting records and apply appropriate compliance labels. For our Remote Information E-Display, this would most likely include:

- Radio frequency compliance
- Electromagnetic compatibility
- Electrical safety
- Battery safety
- Solar charging system safety
- Environmental protection

An external testing laboratory would therefore be required for formal compliance testing.

7.9 Risk Assessment

TABLE XVIII

RISK ASSESSMENT

Risk	Category	Likelihood	Impact	Priority	Mitigation Strategy	Contingency Plan
Insufficient Solar Energy	Technical	Medium	High	High	Power optimisation, energy modelling, increased solar / battery capacity	Increase panel / battery size, reduce display power
Communication Failure	Technical	Medium	High	High	RF range testing, antenna optimisation	Reposition equipment, improve antenna
Water / Environmental Damage	Technical	Medium	High	High	Sealed weather enclosures, environmental testing	Redesign enclosure, improve sealing, change materials
Battery Degradation	Technical	Medium	High	High	Protection system, temperature monitoring	Alternate battery chemistry, redesign charging system
Display Degradation	Technical	Medium	Medium	Medium	Test brightness, visibility and power	Investigate a higher brighter LCD
Software Failure	Technical	Medium	High	High	Error handling, power loss testing	Reduce complexity in code and include fault recovery mechanisms
Component Shortage	Project	Medium	High	High	Identify alternatives during design	Redesign PCB section, stock critical components

Manufacturing delay	Project	Medium	High	Medium	Create realistic lead times or schedule buffers before deadline	Use alternative manufacturer
Budget Overrun	Project	Medium	High	High	Supplier quotes before large orders	Delay non essential work
Testing Delays	Project	Medium	High	High	Book labs early	Fix failed subsystem
Schedule delay	Project	Medium	High	High	Stage the WBS	User alternative site

8. Summary & Conclusion

The project has successfully prototyped a long-range wireless, autonomously charging remote bushfire information display that improves the safety of outdoor recreationists, and benefits both park rangers and emergency authorities.

By utilising an ESP32 microcontroller to control a ILI9488 TFT LCD display via communications over LoRa with an SX1278 module, operators can send bushfire safety data with ease. The system is powered by a multi-stage solar-charged 2S battery system that maintains safe voltage operation via a BMS, and follows safety, EMI and radio broadcast standards.

The product distinguishes itself with a motion-activated wakeup system to minimise power loss, as well as complete network independence with LoRa communication, keeping safety information through outages. It also prioritises an inexpensive deployment to be distributed nationwide for communities of different wealth.

The product uses an IP strategy that allows for deployment without copies being made that infringe on the power supply system. It also keeps the LoRa communication as trade secret to prevent malicious interruptions or usage of non-operators, but keeps hardware open-source to allow community members to create infrastructure that supports installation.

The development posed challenges to the group and led to many design decisions to be made, mostly dealing with the use of LoRa over alternatives such as WiFi and cellular, as well as implementing communication between the sender and receiver devices.

The project has been able to identify further development before the commercial deployment of the product, particularly in long-term environmental testing, reliability, manufacturing and field deployment. Overall, the system provides a practical foundation for a scalable remote bushfire information network, with future development focused on improving reliability, reducing costs and enabling wider deployment across Australia.

Live bushfire safety information should never rely on infrastructure that can be affected by the same bushfires and emergencies they aim to prevent. By utilising radio long-range communication, our product offers a simple, nationwide deployable alternative to major products such as roadside fire rating signs. Our product allows for critical safety information to be distributed to remote locations and hiking trails across all of Australia where pre-existing devices would fail.

9. References

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10. Generative AI Use Disclosure

Generative AI used (select as appropriate): **Yes** / No

Tool Used Purpose Sections Affected

Gemini - Brainstorming: Sections 2, 4, 6

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Name: Christopher Weinstein

zID: z5487950

Signature: 

11. Appendix

11.1 Receiver Code

```
#include <SPI.h>
#include <Adafruit_GFX.h>
#include <TFT_eSPI.h>
#include <LoRa.h>
#include "driver/rtc_io.h"

// Wake-up Bitmask: GPIO 26 (LoRa DIO0) and GPIO 34 (PIR)
#define WAKEUP_BITMASK ((1ULL << GPIO_NUM_26) | (1ULL << GPIO_NUM_34))

#define LORA_SS      5
#define LORA_RST     32
#define LORA_DIO0    26

#define SPI_SCK      18
#define SPI_MISO     19
#define SPI_MOSI     23

TFT_eSPI tft = TFT_eSPI();
#define PIR 34
#define ILI_LED 21

enum FireRating {
    NONE = 0,
    MODERATE = 1,
    HIGH_FIRE = 2,
    EXTREME = 3,
    CATASTROPHIC = 4
};

struct BushfirePacket {
    uint8_t fireRating;
    uint8_t trailClosed;
    uint8_t fireBan;
    uint16_t direction;
    uint16_t distance;
};

BushfirePacket packet;

// preserve display state across sleep
```

```

RTC_DATA_ATTR FireRating rating = MODERATE;
RTC_DATA_ATTR bool fireBan = false;
RTC_DATA_ATTR bool trailClosed = false;
RTC_DATA_ATTR uint16_t direction = 0;
RTC_DATA_ATTR uint16_t distance = 100;

const int SLEEPTIME = 10000;
int state = 2;
unsigned long fallTime = 0;

String directionToString(int angle);
String ratingToString(FireRating r);
uint16_t ratingColour();
void drawArrow(int angle);
void drawScreen();

void setup() {
    Serial.begin(115200);
    delay(200);

    rtc_gpio_hold_dis((gpio_num_t)LORA_RST);

    pinMode(PIR, INPUT);
    pinMode(ILI_LED, OUTPUT);
    digitalWrite(ILI_LED, HIGH);

    pinMode(LORA_SS, OUTPUT);
    digitalWrite(LORA_SS, HIGH);
    pinMode(LORA_RST, OUTPUT);
    digitalWrite(LORA_RST, HIGH);

    SPI.begin(SPI_SCK, SPI_MISO, SPI_MOSI, LORA_SS);

    // prevent accidental reset
    LoRa.setPins(LORA_SS, -1, LORA_DIO0);
    if (!LoRa.begin(433E6)) {
        Serial.println("[ERROR] LoRa soft initialization failed!");
        while (1);
    }

    // get and print wakeup reason
    esp_sleep_wakeup_cause_t wakeup_reason =
    esp_sleep_get_wakeup_cause();

```

```

if (wakeup_reason == ESP_SLEEP_WAKEUP_EXT1) {
    uint64_t status = esp_sleep_get_ext1_wakeup_status();

    if (status & (1ULL << GPIO_NUM_26)) {
        Serial.println("Woken up by LoRa DIO0! Reading packet from
FIFO...");

        int packetSize = LoRa.parsePacket();
        if (packetSize == sizeof(BushfirePacket)) {
            LoRa.readBytes((uint8_t *)&packet, sizeof(packet));

            rating = (FireRating)packet.fireRating;
            trailClosed = packet.trailClosed;
            fireBan = packet.fireBan;
            direction = packet.direction;
            distance = packet.distance;

            Serial.println(">>> Wake-up packet successfully
decoded!");
        } else {
            Serial.printf(">>> Wake-up packet size mismatch or
empty: %d bytes\n", packetSize);
        }
    } else if (status & (1ULL << GPIO_NUM_34)) {
        Serial.println("Woken up by PIR motion sensor.");
    }
} else {
    Serial.println("Standard Power-On / Cold Reset.");
}

// re-allocate reset pin
LoRa.setPins(LORA_SS, LORA_RST, LORA_DIO0);

tft.init();
tft.setRotation(0);
drawScreen();

LoRa.receive();
fallTime = millis();
}

void loop() {

```

```

// update motion sensor reading
int motion = digitalRead(PIR);

if (motion == HIGH && state != 1) {
    digitalWrite(ILI_LED, HIGH);
    state = 1;
}

if (motion == LOW && state == 1) {
    fallTime = millis();
    state = 2;
}

// LoRa receive
int packetSize = LoRa.parsePacket();

if (packetSize == sizeof(BushfirePacket)) {
    LoRa.readBytes((uint8_t *)&packet, sizeof(packet));

    rating = (FireRating)packet.fireRating;
    trailClosed = packet.trailClosed;
    fireBan = packet.fireBan;
    direction = packet.direction;
    distance = packet.distance;
    drawScreen();

    fallTime = millis();
}

if (state == 2 && (millis() - fallTime >= SLEEPTIME)) {
    // keep LoRa receiving
    LoRa.receive();

    digitalWrite(ILI_LED, LOW);
    delay(100);

    // wakeup with motion sensor or LoRa
    esp_sleep_enable_ext1_wakeup(WAKEUP_BITMASK,
ESP_EXT1_WAKEUP_ANY_HIGH);

    // keep LoRa on (don't reset)
    gpio_deep_sleep_hold_en();
    rtc_gpio_hold_en((gpio_num_t)LORA_RST);
}

```

```

        // eepy
        esp_deep_sleep_start();
    }
}

String ratingToString(FireRating r) {
    switch(r) {
        case NONE:            return "None";
        case MODERATE:        return "Moderate";
        case HIGH_FIRE:       return "High";
        case EXTREME:         return "Extreme";
        case CATASTROPHIC:    return "Catastrophic";
        default:              return "Unknown";
    }
}

uint16_t ratingColour() {
    switch(rating) {
        case NONE:            return TFT_GREEN;
        case MODERATE:        return TFT_YELLOW;
        case HIGH_FIRE:       return TFT_ORANGE;
        case EXTREME:         return TFT_RED;
        case CATASTROPHIC:    return TFT_MAGENTA;
        default:              return TFT_WHITE;
    }
}

void drawArrow(int angle) {
    const int cx = 160;
    const int cy = 370;
    const int L = 45;
    const int headLength = 20;

    float rad = angle * PI / 180.0;
    int x2 = cx + L * sin(rad);
    int y2 = cy - L * cos(rad);

    tft.drawLine(cx, cy, x2, y2, ratingColour());

    float left  = rad + 150 * PI / 180.0;
    float right = rad - 150 * PI / 180.0;

```

```

int hx1 = x2 + headLength * sin(left);
int hy1 = y2 - headLength * cos(left);
int hx2 = x2 + headLength * sin(right);
int hy2 = y2 - headLength * cos(right);

tft.fillTriangle(x2, y2, hx1, hy1, hx2, hy2, ratingColour());
tft.fillCircle(cx, cy, 3, TFT_WHITE);

String distText;
if (distance >= 1000) {
    if (distance % 1000 == 0)
        distText = String(distance / 1000) + " km";
    else
        distText = String(distance / 1000.0, 1) + " km";
} else {
    distText = String(distance) + " m";
}

String info = directionToString(angle) + " " + distText;

tft.setTextColor(TFT_CYAN, TFT_BLACK);
tft.drawCentreString(info, cx, 265, 2);
}

void drawScreen() {
    tft.fillScreen(TFT_BLACK);

    // Fire Rating Title
    tft.setTextFont(4);
    tft.drawCentreString("Fire Rating", 160, 5, 2);

    // Rating Box
    tft.fillRoundRect(60, 30, 200, 45, 8, ratingColour());
    tft.setTextColor(TFT_BLACK, ratingColour());
    tft.setTextDatum(MC_DATUM);
    tft.drawCentreString(ratingToString(rating), 160, 43, 4);
    tft.setTextDatum(TL_DATUM);

    // Fire Ban Box
    if (fireBan) {
        tft.fillRoundRect(60, 95, 200, 45, 8, TFT_RED);
        tft.setTextColor(TFT_WHITE, TFT_RED);
        tft.drawCentreString("TOTAL FIRE BAN", 160, 108, 4);
    }
}

```

```

}

// Trail Closed Box
if (trailClosed) {
    tft.fillRoundRect(60, 155, 200, 45, 8, TFT_ORANGE);
    tft.setTextTextColor(TFT_BLACK, TFT_ORANGE);
    tft.drawCentreString("Trail Closed", 160, 168, 4);
}

// Compass Title
tft.setTextTextColor(TFT_WHITE, TFT_BLACK);
tft.drawCentreString("Fire Assembly Area", 160, 240, 4);

// Compass Circle
const int cx = 160;
const int cy = 370;
const int radius = 58;
tft.drawCircle(cx, cy, radius, ratingColour());

// Compass Ticks
for (int a = 0; a < 360; a += 22) {
    float r = a * PI / 180.0;
    int x1 = cx + (radius - 7) * sin(r);
    int y1 = cy - (radius - 7) * cos(r);
    int x2 = cx + radius * sin(r);
    int y2 = cy - radius * cos(r);

    tft.drawLine(x1, y1, x2, y2, TFT_DARKGREY);
}

// Compass Labels
tft.setTextTextColor(TFT_WHITE, TFT_BLACK);
tft.drawCentreString("N", cx, cy - radius - 18, 2);
tft.drawCentreString("S", cx, cy + radius + 4, 2);
tft.drawCentreString("E", cx + radius + 12, cy - 6, 2);
tft.drawCentreString("W", cx - radius - 18, cy - 6, 2);

drawArrow(direction);
}

String directionToString(int angle) {
    switch(angle) {
        case 0:    return "North";
    }
}

```

```
case 22: return "North-North-East";
case 45: return "North-East";
case 67: return "East-North-East";
case 90: return "East";
case 112: return "East-South-East";
case 135: return "South-East";
case 157: return "South-South-East";
case 180: return "South";
case 202: return "South-South-West";
case 225: return "South-West";
case 247: return "West-South-West";
case 270: return "West";
case 292: return "West-North-West";
case 315: return "North-West";
case 337: return "North-North-West";
default: return "Unknown";
}
}
```

11.2 Sender Code

```
#include <SPI.h>
#include <WiFi.h>
#include <WebServer.h>
#include <LoRa.h>

const char webpage[] PROGMEM = R"rawliteral(
<!DOCTYPE html>
<html>

<head>

<meta charset="UTF-8">

<title>Bushfire Warning</title>

<style>

body{
    font-family:Arial;
    text-align:center;
    margin-top:50px;
    background:#f5f5f5;
}

h1{
    font-size:36px;
}

h2{
    margin-top:30px;
}

select{
```

```

    font-size:20px;
    padding:10px;
    width:220px;
}

button{
    padding:15px 25px;
    font-size:20px;
    margin-top:25px;
    cursor:pointer;
}

label{
    font-size:20px;
}

</style>

</head>

<body>

<h1>Bushfire Warning Control</h1>

<h2>Fire Rating</h2>

<select id="rating">

<option value="0">None</option>
<option value="1">Moderate</option>
<option value="2">High</option>
<option value="3">Extreme</option>
<option value="4">Catastrophic</option>

```

```
</select>

<br><br>

<label>

<input type="checkbox" id="trail">

Trail Closed

</label>

<br><br>

<label>

<input type="checkbox" id="ban">

Total Fire Ban

</label>

<br><br>

<h2>Fire Direction</h2>

<select id="direction">
```

```
<option value="0">North</option>
<option value="22">North-North-East</option>
<option value="45">North-East</option>
<option value="67">East-North-East</option>
<option value="90">East</option>
<option value="112">East-South-East</option>
<option value="135">South-East</option>
<option value="157">South-South-East</option>
<option value="180">South</option>
<option value="202">South-South-West</option>
<option value="225">South-West</option>
<option value="247">West-South-West</option>
<option value="270">West</option>
<option value="292">West-North-West</option>
<option value="315">North-West</option>
<option value="337">North-North-West</option>
```

```
</select>
```

```
<br><br>
```

```
<h2>Distance To Assembly Area</h2>
```

```
<input
type="number"
id="distance"
value="100"
min="0"
max="5000"
step="10">
```

```
<span> metres</span>
```

```
<br><br>
```

```
<button onclick="sendData()">
```

Broadcast Sign

```
</button>
```

```
<script>
```

```
function sendData(){
```

```
    const rating =  
    document.getElementById("rating").value;
```

```
    const trail =  
    document.getElementById("trail").checked ? 1 : 0;
```

```
    const ban =  
    document.getElementById("ban").checked ? 1 : 0;
```

```
    const direction =  
    document.getElementById("direction").value;
```

```
    const distance =  
    document.getElementById("distance").value;
```

```
    fetch("/update?rating="+rating+  
    "&trail="+trail+  
    "&ban="+ban+  
    "&direction="+direction+  
    "&distance="+distance)
```

```

.then(response => response.text())

.then(text => {

alert(text);

}))

.catch(error => {

alert("Transmission Failed");

console.log(error);

});

}

</script>

</body>

</html>
)rawliteral";
// LoRa Pins (ESP32-Wroom-32E)

#define LORA_SS      5
#define LORA_RST     32
#define LORA_DIO0    26

```

```

// SPI Pins (ESP32-Wroom-32E)

#define SPI_SCK      18
#define SPI_MISO     16
#define SPI_MOSI     23

// WiFi

const char* ssid = "BushfireSign";
const char* password = "12345678";

WebServer server(80);

// Packet Structure

struct BushfirePacket
{
    uint8_t fireRating;
    uint8_t trailClosed;
    uint8_t fireBan;
    uint16_t direction;
    uint16_t distance;
};

BushfirePacket packet = {0,0,0,0,100};

// Setup

void setup()
{
    Serial.begin(9600);
    delay(3000);

```

```

Serial.println();
Serial.println("TRANSMITTER START");

// SPI

Serial.println("Starting SPI");

SPI.begin(
    SPI_SCK,
    SPI_MISO,
    SPI_MOSI
);

Serial.println("Starting LoRa");

LoRa.setPins(
    LORA_SS,
    LORA_RST,
    LORA_DIO0
);

if (!LoRa.begin(434E6))
{
    Serial.println("LoRa FAILED");
    while(true);
}
else
{
    Serial.println("LoRa READY");
}

// WiFi Access Point

```

```

Serial.println("Starting WiFi");

if (WiFi.softAP(ssid, password))
{
    Serial.println("WiFi Started");
}
else
{
    Serial.println("WiFi FAILED");
}

Serial.print("IP Address: ");
Serial.println(WiFi.softAPIP());

// Web Server

server.on("/", HTTP_GET, [] ()
{
    Serial.println("Browser connected");

    server.send_P(
        200,
        "text/html",
        webpage
    );

});

// Update Route

server.on("/update", HTTP_GET, [] ()
{
    Serial.println("Update received");
}

```

```

if (!server.hasArg("rating") ||
    !server.hasArg("trail") ||
    !server.hasArg("ban") ||
    !server.hasArg("direction") ||
    !server.hasArg("distance"))
{
    server.send(
        400,
        "text/plain",
        "Missing parameters"
    );

    return;
}

packet.fireRating = server.arg("rating").toInt();
packet.trailClosed = server.arg("trail").toInt();
packet.fireBan     = server.arg("ban").toInt();
packet.direction = server.arg("direction").toInt();
packet.distance = server.arg("distance").toInt();

Serial.println("Sending packet");

Serial.print("Rating: ");
Serial.println(packet.fireRating);

Serial.print("Trail: ");
Serial.println(packet.trailClosed);

Serial.print("Ban: ");
Serial.println(packet.fireBan);

Serial.print("Direction: ");
Serial.println(packet.direction);

```

```

        Serial.print("Distance: ");
        Serial.print(packet.distance);
        Serial.println(" m");

        if (LoRa.beginPacket() == 0)
        {
            Serial.println("LoRa begin failed");
        }
        else
        {
            LoRa.write(
                (uint8_t*)&packet,
                sizeof(packet)
            );

            int result = LoRa.endPacket();

            if(result == 1)
            {
                Serial.println("LoRa TX success");
            }
            else
            {
                Serial.println("LoRa TX failed");
            }
        }

        Serial.println("Packet sent");
        server.send(
            200,
            "text/plain",
            "Broadcast successful"
        );
    });

    // Start Server

```

```
server.begin();

Serial.println("SERVER STARTED");
Serial.println("Waiting for browser...");
}

void loop()
{
  server.handleClient();
}
```

11.3 PCB BOM

Item	Part Number	Manufacturer	Quantity	Unit Cost	Cost	Link
LDO	TLV73333 PQDBVRQ 1	Texas Instruments	1	\$0.77	\$0.77	https://au.mouser.com/ProductDetail/Texas-Instruments/TLV73333PQDBVRQ1?qs=9r4v7xj2LnnYANXVL3PB4Q%3D%3D
1uF cap	KGM21AR 51E105KL	Kyocera	4	\$0.26	\$1.04	https://au.mouser.com/ProductDetail/KYOCERA-AVX/KGM21AR51E105KL?qs=sGAepiMZZMuMW9TILBQkXqLYvz3DzaXfwIvgK7bVzak%3D
Buck	LMR51610 XFQDBVR Q1	Texas Instruments	2	\$2.76	\$5.52	https://au.mouser.com/ProductDetail/Texas-Instruments/LMR51610XFQDBVRQ1?qs=iLKYxqzqNS748iUg0IV17HA%3D%3D

33uH inductor	SRN8040HA-330M	Bourns	2	\$0.93	\$1.86	https://au.mouser.com/ProductDetail/Bourns/SRN8040HA-330M?qs=2SLPxufLcgBRMUyV6ne5Ew%3D%3D
22.1kOhm resistor	ERJ-UP6F2 212V	Panasonic	2	\$0.41	\$0.82	https://au.mouser.com/ProductDetail/Panasonic/ERJ-UP6F2212V?qs=sGAEpiMZZMtG0KNrPCHnjRpfexHZHoPIXlAJksbxLxHIWZy0of5Dsw%3D%3D
118kOhm resistor	ERA-6AEB 1183V	Panasonic	1	\$0.15	\$0.15	https://au.mouser.com/ProductDetail/Panasonic/ERA-6AEB1183V?qs=sGAEpiMZZMtG0KNrPCHnjfgxrTtCmcjiS84CbomSrLc%3D

220kOhm resistor	ERJ-S06J2 24V	Panasonic	1	\$0.48	\$0.48	https://au.mouser.com/ProductDetail/Panasonic/ERJ-S06J224V?qs=sGAEpiMZZMtG0KNrPCHnjXA8RZptup0%2FvTbIuqWH1UU%3D
2.2uF cap	GRT21BD7 2A225KE1 3L	Murata	8	\$0.89	\$7.12	https://au.mouser.com/ProductDetail/Murata-Electronics/GRT21BD72A225KE13L?qs=ulEaXIWI0c%2BmG%2FIhw%2FxmGA%3D%3D
22uF cap	GRT21BR6 1E226ME1 3L	Murata	8	\$0.67	\$5.36	https://au.mouser.com/ProductDetail/Murata-Electronics/GRT21BR61E226ME13L?qs=drGMNd%2BkGPNuWqerICGuZg%3D%3D

0.1u/100nF cap	885012207 098	Würth	6	\$0.15	\$0.90	https://au.mouser.com/ProductDetail/Wurth-Elektronik/885012207098?qs=0KOYDY2FL29sZCNyRsrK9g%3D%3D
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Total: \$24.02